

On the noise estimation statistics

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ABSTRACT

Learning with noisy labels has attracted much attention during the past few decades. A fundamental problem is how to estimate noise proportions from corrupted data. Previous studies on this issue resort to the estimations of class distributions, conditional distributions, or the kernel embedding of distributions. In this paper, we present another simple and effective approach for noise estimation. The basic idea is to utilize the first- and second-order statistics of observed data, and the positive semi-definiteness of covariance matrices. Then, an upper bound on noise estimation is provided without additional assumptions over data distribution. Based on this idea and using the locality property of random noise, we develop the *Noise Estimation Statistics with Clusters (NESC)* method, which firstly clusters the corrupted data by *k*-means algorithm, and then makes noise estimation from clusters based on the first- and second-order statistics. We present the existence, uniqueness and convergence analysis of our noise estimation, and empirical studies verify the effectiveness of the NESC method.

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1. Introduction

In many real applications, we observe samples whose labels are corrupted in some way from the ground truth, e.g., a document may be mis-classified manually due to human error or bias, and a doctor may make incorrect diagnoses to patients due to his knowledge and experience. Crowdsourcing has been an efficient way to gather a large amount of data on the Internet yet with inaccurate and corrupted labels [9]. In the positive and unlabeled learning, we have access to some positive and unlabeled samples, but with a lack of negative samples [18,21,20].

In this work, we focus on binary classification in the presence of random noise, where each observed label has been flipped independently with a certain probability instead of seeing the true label, and training samples of each class are contaminated from the other class. This random noise model was firstly introduced by Angluin and Laird [1], and motivated a series of follow-up studies in learning theory. Aslam and Decatur [2] and Cesa-Bianchi et al. [13] made finite VC-dimension to characterize the learnability of random noise model. For online mistake bound model, Ben-David et al. [4] studied the learnability based on Littlestone dimension. Kearns [28,29] proposed the famous statistical query model by capturing statistical properties rather than individual example. Bshouty et al. [11] gave a statistical query algorithm in noise-tolerant and distribution-free classification. Kalai and Servedio [27] presented theoretical analysis on boosting algorithms in the presence of random noise.

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Generally speaking, noisy labels may disturb the learning process and deteriorate the effectiveness of learning algorithms, e.g., random noise defeats all convex potential boosters [33], and support vector machines (SVMs) tend to overfit on noisy labels [34,46]. Numerous practical algorithms have been developed for learning with noisy labels, including outlier detections [5,6,10,31], re-weights of training examples [42,45], perceptron-style algorithms [12,14–16,19], boosting algorithms [23,30], robust loss based algorithms [17,25,34,35,43,47]. The interested readers are referred to surveys [22,38].

Noise estimation has been one of the fundamental problems for learning with noisy labels, which is helpful to design some applicable learning algorithms; for example, it is feasible to deal with noisy labels by modifying the biased or cost-sensitive losses under the knowledge of random noise proportions [24,37,39]. For noise estimation, Scott et al. [44] formalized it as the mutually contaminated distributions framework, and estimated noise proportions from mixture proportion distributions based on an irreducibility assumption over data distribution. Ramaswamy et al. [41] introduced another algorithm for mixture proportion distributions via kernel embedding of distributions, and presented the convergence analysis under an assumption which correlates data distribution with kernel functions. The conditional probability w.r.t. corrupted data distribution has also been suggested for noise estimation [32,36].

This work develops another simple and effective method for noise estimation, and the main contributions can be summarized as follows:

- An upper bound on noise estimation is provided without additional assumptions over data distribution. The basic idea is to utilize the first- and second-order statistics of observed data, as well as the positive semi-definiteness of covariance matrices.
- We prove the existence and uniqueness of our noise estimation, and present the convergence analysis without additional assumptions over data distribution. Our theoretical analysis is also independent of any hypothesis space and kernel function.
- Based on theoretical analysis, we develop the *Noise Estimation Statistics with Clusters (NESC)* method, which firstly clusters the corrupted data by k -means algorithm, and then performs noise estimation from clusters based on the first- and second-order statistics.
- Empirical studies show that our NESC achieves significantly better performance than the state-of-the-art algorithms on noise estimation, also with the lowest running-time cost.

The rest of this paper is structured as follows. Section 2 presents some preliminaries. Section 3 introduces previous methods on noise estimations. Section 4 gives an upper bound on noise estimation. Section 5 shows the convergence analysis. Section 6 proposes the NESC method. Section 7 conducts some empirical studies. Section 8 concludes with future work. All detailed proofs are presented in the Appendix.

2. Preliminaries

Let $\mathcal{X} \subseteq \mathbb{R}^d$ and $\mathcal{Y} = \{+1, -1\}$ denote the instance and label space, respectively. Assume that D is an underlying (unknown) distribution over the product space $\mathcal{X} \times \mathcal{Y}$, and $D_{\mathcal{X}}$ represents the corresponding marginal distribution over the instance space $\mathcal{X}$. We denote by

$$\eta(\mathbf{x}) = \Pr[y = +1|\mathbf{x}] \quad \text{and} \quad \pi = \Pr[y = +1]$$

the conditional probability and ratio of positive instances w.r.t. distribution D , respectively. We further express distribution D in terms of the ratio π and class distributions P and Q , as mentioned by Menon et al. [36], where

$$P(\mathbf{x}) = \Pr[\mathbf{x}|y = +1] \quad \text{and} \quad Q(\mathbf{x}) = \Pr[\mathbf{x}|y = -1].$$

Let $f_P(\mathbf{x})$, $f_Q(\mathbf{x})$, and $f_{D_{\mathcal{X}}}(\mathbf{x})$ denote the probability densities of distributions P , Q and $D_{\mathcal{X}}$, respectively, and we consider the Lebesgue measure for all density functions in this work.

Given class distribution P , we define the *first- and second-order statistics*, respectively, as

$$\mu_P = E_{\mathbf{x} \sim P}[\mathbf{x}] \quad \text{and} \quad \Sigma_P = E_{\mathbf{x} \sim P}[\mathbf{x}\mathbf{x}^\top],$$

which are also called the mean and second-moment (un-centered covariance) matrix of random variable $\mathbf{x}$ drawn from class distribution P . This gives the covariance matrix

$$\text{Cov}_P = \Sigma_P - \mu_P \mu_P^\top.$$

We also define the first- and second-order statistics w.r.t. class distribution Q , respectively, as

$$\mu_Q = E_{\mathbf{x} \sim Q}[\mathbf{x}] \quad \text{and} \quad \Sigma_Q = E_{\mathbf{x} \sim Q}[\mathbf{x}\mathbf{x}^\top].$$

Suppose that there is a training sample $S_n = \{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_n, y_n)\}$, where each element is drawn independently and identically (i.i.d.) according to distribution D . In the random noise model, each observed label has been flipped independently with a certain probability instead of seeing the true label [1]. Specifically, each label y_i is corrupted by random noise with proportions τ_+ and τ_- , that is,

$$\tau_+ = \Pr[\tilde{y}_i = -1 | y_i = +1] \quad \text{and} \quad \tau_- = \Pr[\tilde{y}_i = +1 | y_i = -1].$$

Following [8], we assume $\tau_+ + \tau_- \leq 1$ in this work. In the random noise model, what we can observe is essentially a corrupted training sample

$$\tilde{S}_n = \{(\mathbf{x}_1, \tilde{y}_1), (\mathbf{x}_2, \tilde{y}_2), \dots, (\mathbf{x}_n, \tilde{y}_n)\},$$

where each label has been flipped independently with probability τ_+ or τ_- . The goal of this work is to estimate noise proportions τ_+ and τ_- from the corrupted training sample $\tilde{S}_n$.

Let $\tilde{D}$ be the corrupted distribution, that is, the ground-truth distribution D is corrupted by random noise with proportions τ_+ and τ_- . Denote by

$$\tilde{\eta}(\mathbf{x}) = \Pr[\tilde{y} = +1 | \mathbf{x}] \quad \text{and} \quad \tilde{\pi} = \Pr[\tilde{y} = +1]$$

the conditional probability and ratio of positive instances w.r.t. the corrupted distribution $\tilde{D}$, respectively. We further represent the corrupted distribution $\tilde{D}$ in terms of the ratio $\tilde{\pi}$ and class distributions $\tilde{P}$ and $\tilde{Q}$, where

$$\tilde{P}(\mathbf{x}) = \Pr[\mathbf{x} | \tilde{y} = +1] \quad \text{and} \quad \tilde{Q}(\mathbf{x}) = \Pr[\mathbf{x} | \tilde{y} = -1].$$

We denote by the first- and second-order statistics of distributions $\tilde{P}$ and $\tilde{Q}$, respectively, as

$$\mu_{\tilde{P}} = E_{\mathbf{x} \sim \tilde{P}}[\mathbf{x}], \quad \mu_{\tilde{Q}} = E_{\mathbf{x} \sim \tilde{Q}}[\mathbf{x}], \quad \Sigma_{\tilde{P}} = E_{\mathbf{x} \sim \tilde{P}}[\mathbf{x}\mathbf{x}^\top], \quad \Sigma_{\tilde{Q}} = E_{\mathbf{x} \sim \tilde{Q}}[\mathbf{x}\mathbf{x}^\top].$$

We denote by $A > 0$ (resp. $A \geq 0$) if A is a positive definite (resp. semi-definite) matrix, and $\lambda_{\min}(A)$ represents the minimal eigenvalue of matrix A . Given a set $\mathcal{A}$, let $|\mathcal{A}|$ denote its cardinality. We also write $[n] := \{1, 2, \dots, n\}$ for an integer $n > 0$.

3. Previous studies on noise estimation

We first present the following relations for density functions in the noise and noise-free settings, and the detailed proof is presented in Appendix A.

Lemma 1. If $f_P(\mathbf{x})$, $f_Q(\mathbf{x})$, $f_{\tilde{P}}(\mathbf{x})$, $f_{\tilde{Q}}(\mathbf{x})$ and $f_{D_{\mathcal{X}}}(\mathbf{x})$ are density functions of distributions P , Q , $\tilde{P}$, $\tilde{Q}$ and $D_{\mathcal{X}}$, respectively, with noise proportions τ_+ and τ_- and $\tilde{\pi} = \Pr[\tilde{y} = +1]$, then we have, for $\tilde{\pi} \in (0, 1)$,

$$f_{D_{\mathcal{X}}}(\mathbf{x}) = \tilde{\pi} f_{\tilde{P}}(\mathbf{x}) + (1 - \tilde{\pi}) f_{\tilde{Q}}(\mathbf{x}), \quad (1)$$

$$f_{\tilde{P}}(\mathbf{x}) = \frac{\tilde{\pi} - \tau_-}{\tilde{\pi}(1 - \tau_-)} f_P(\mathbf{x}) + \frac{\tau_-(1 - \tilde{\pi})}{\tilde{\pi}(1 - \tau_-)} f_Q(\mathbf{x}), \quad (2)$$

$$f_{\tilde{Q}}(\mathbf{x}) = \frac{1 - \tilde{\pi} - \tau_+}{(1 - \tilde{\pi})(1 - \tau_+)} f_Q(\mathbf{x}) + \frac{\tau_+ \tilde{\pi}}{(1 - \tilde{\pi})(1 - \tau_+)} f_P(\mathbf{x}); \quad (3)$$

and we also have $\tau_- \in [0, \tilde{\pi}]$ and $\tau_+ \in [0, 1 - \tilde{\pi}]$ for $\tilde{\pi} \in [0, 1]$, which yields that $\tau_- = 0$ if $\tilde{\pi} = 0$, and $\tau_+ = 0$ if $\tilde{\pi} = 1$.

Lemma 1 can be viewed as another expression of contaminated distributions of random noise model [44]. Based on this lemma, we can observe that noise estimation is not a well-defined problem, as mentioned in [7,44], which can be shown by the following example:

Example 1. Suppose that $\tilde{P}$ and $\tilde{Q}$ are the corrupted class distributions from the ground-truth class distributions P and Q with noise proportions τ_- and τ_+ . Let $\tilde{\pi} = \Pr[\tilde{y} = +1]$. We have

$$\tilde{P} = \frac{\tilde{\pi} - \tau_-}{\tilde{\pi}(1 - \tau_-)} P + \frac{(1 - \tilde{\pi})\tau_-}{\tilde{\pi}(1 - \tau_-)} \tilde{Q} \quad \text{and} \quad \tilde{Q} = \frac{1 - \tilde{\pi} - \tau_+}{(1 - \tilde{\pi})(1 - \tau_+)} Q + \frac{\tau_+ \tilde{\pi}}{(1 - \tilde{\pi})(1 - \tau_+)} \tilde{P}.$$

We can also rewrite distributions $\tilde{P}$ and $\tilde{Q}$ as

$$\tilde{P} = \frac{\tilde{\pi} - \tau_-/2}{\tilde{\pi}(1 - \tau_-/2)} P' + \frac{(1 - \tilde{\pi})\tau_-/2}{\tilde{\pi}(1 - \tau_-/2)} \tilde{Q} \quad \text{and} \quad \tilde{Q} = \frac{1 - \tilde{\pi} - \tau_+/2}{(1 - \tilde{\pi})(1 - \tau_+/2)} Q' + \frac{\tau_+ \tilde{\pi}/2}{(1 - \tilde{\pi})(1 - \tau_+/2)} \tilde{P}$$

where

$$P' = \frac{(2 - \tau_-)(\tilde{\pi} - \tau_-)}{(1 - \tau_-)(2\tilde{\pi} - \tau_-)}P + \frac{\tau_-(1 - \tilde{\pi})}{(1 - \tau_-)(2\tilde{\pi} - \tau_-)}\tilde{Q},$$

$$Q' = \frac{(1 - \tilde{\pi} - \tau_+)(2 - \tau_+)}{(1 - \tau_+)(2 - 2\tilde{\pi} - \tau_+)}Q + \frac{\tau_+\tilde{\pi}}{(1 - \tau_+)(2 - 2\tilde{\pi} - \tau_+)}\tilde{P}.$$

Thus, we can also view the corrupted distributions $\tilde{P}$ and $\tilde{Q}$ from the ground-truth distributions P' and Q' with noise proportions $\tau_-/2$ and $\tau_+/2$.

Example 1 shows that the corrupted class distributions can be viewed as different ground-truth class distributions with different noise proportions, and we can not decide which estimation is correct without additional information; therefore, noise estimation is not a well-defined problem.

To make the problem well-defined, Scott et al. [44] introduced the mutually contaminated distribution framework, and made noise estimation from mixture proportion distributions based on an irreducibility assumption over data distribution. Specifically, the noise proportions are estimated (written with our notations) by

$$\tau_-^* = \max \left\{ \tau \in [0, \tilde{\pi}] : \exists \text{ distribution } P \text{ s.t. } \tilde{P} = \frac{\tilde{\pi} - \tau}{\tilde{\pi}(1 - \tau)}P + \frac{\tau(1 - \tilde{\pi})}{\tilde{\pi}(1 - \tau)}\tilde{Q} \right\},$$

$$\tau_+^* = \max \left\{ \tau \in [0, 1 - \tilde{\pi}] : \exists \text{ distribution } Q \text{ s.t. } \tilde{Q} = \frac{1 - \tilde{\pi} - \tau}{(1 - \tilde{\pi})(1 - \tau)}Q + \frac{\tau\tilde{\pi}}{(1 - \tilde{\pi})(1 - \tau)}\tilde{P} \right\},$$

where the ratio $\tilde{\pi}$ and class distributions $\tilde{Q}$ and $\tilde{P}$ can be estimated from the corrupted training sample $\tilde{S}_n$. Returning to Example 1, Scott et al. [44] would make noise estimation at least τ_- and τ_+ , rather than $\tau_-/2$ and $\tau_+/2$.

To avoid estimating the corrupted class distributions $\tilde{Q}$ and $\tilde{P}$, Ramaswamy et al. [41] proposed to embed distributions $\tilde{Q}$ and $\tilde{P}$ into a Reproduced Kernel Hilbert Space (RKHS), and solve this problem by convex quadratic programming. Also, they presented the convergence analysis under an assumption (written with our notations below), which correlates the corrupted class distributions $\tilde{Q}$ and $\tilde{P}$ with kernel function κ ; in particular, there is an h in the unit norm ball of the RKHS such that

$$E_{X \sim \tilde{Q}}[h(X)] \leq \inf_x [h(x)] + \beta \leq E_{X \sim \tilde{P}}[h(X)] - \alpha$$

for some margin $\alpha > 0$ and tolerance β . Obviously, this method depends on the suitability between kernel function and class distributions $\tilde{Q}$ and $\tilde{P}$.

Liu and Tao [32] and Menon et al. [36] made noise estimation based on the minimum and maximum of conditional probability of positive instances w.r.t. corrupted distribution $\tilde{D}$, that is

$$\tau_-^* = \min_{\mathbf{x} \in \mathcal{X}} \{\hat{\eta}(\mathbf{x})\} \quad \text{and} \quad \tau_+^* = 1 - \max_{\mathbf{x} \in \mathcal{X}} \{\hat{\eta}(\mathbf{x})\}$$

where the conditional probability $\hat{\eta}(\mathbf{x})$ can be estimated from the corrupted training sample $\tilde{S}_n$.

In summary, previous studies resort to the estimations of class distributions or conditional distributions, or embedding class distributions into a RKHS, and some additional assumptions are made necessarily for those estimations of distributions and embeddings from the corrupted training data $\tilde{S}_n$, which are unknown for real applications.

4. An upper bound on noise estimation

Based on Lemma 1, we could present some relations for data statistics in the noise and noise-free settings, and the detailed proof is presented in Appendix B.

Lemma 2. If $\mu_P, \mu_Q, \mu_{\tilde{P}}, \mu_{\tilde{Q}}, \Sigma_P, \Sigma_Q, \Sigma_{\tilde{P}}$ and $\Sigma_{\tilde{Q}}$ are the first- and second-order statistics of distributions $P, Q, \tilde{P}$ and $\tilde{Q}$, respectively, with noise proportions τ_+ and τ_- and $\tilde{\pi} = \Pr[\tilde{y} = +1]$, then we have

$$\mu_P = \frac{\tilde{\pi}(1 - \tau_-)}{\tilde{\pi} - \tau_-}\mu_{\tilde{P}} - \frac{\tau_-(1 - \tilde{\pi})}{\tilde{\pi} - \tau_-}\mu_{\tilde{Q}}, \quad (4)$$

$$\Sigma_P = \frac{\tilde{\pi}(1 - \tau_-)}{\tilde{\pi} - \tau_-}\Sigma_{\tilde{P}} - \frac{\tau_-(1 - \tilde{\pi})}{\tilde{\pi} - \tau_-}\Sigma_{\tilde{Q}}, \quad (5)$$

$$\mu_Q = \frac{(1 - \tilde{\pi})(1 - \tau_+)}{1 - \tilde{\pi} - \tau_+}\mu_{\tilde{Q}} - \frac{\tilde{\pi}\tau_+}{1 - \tilde{\pi} - \tau_+}\mu_{\tilde{P}}, \quad (6)$$

$$\Sigma_Q = \frac{(1 - \tilde{\pi})(1 - \tau_+)}{1 - \tilde{\pi} - \tau_+}\Sigma_{\tilde{Q}} - \frac{\tilde{\pi}\tau_+}{1 - \tilde{\pi} - \tau_+}\Sigma_{\tilde{P}}. \quad (7)$$

Notice that the converse does not hold in Lemma 2; in other words, we can not guarantee that μ_P , μ_Q , Σ_P and Σ_Q are statistics of some distributions P and Q even if Eqns. (4)-(7) hold, which can be shown as follows:

Example 2. Let $\tilde{\pi} = 1/2$ and $\Sigma_{\tilde{P}} = \Sigma_{\tilde{Q}}/3$ with $\Sigma_{\tilde{P}} \succ 0$. The ground-truth noise τ_- is unknown, and if we assume $\tau_- = 1/3 < \tilde{\pi}$, then we have, according to Eqn. (5),

$$\Sigma_P = \frac{\tilde{\pi}(1-\tau_-)}{\tilde{\pi}-\tau_-}\Sigma_{\tilde{P}} - \frac{\tau_-(1-\tilde{\pi})}{\tilde{\pi}-\tau_-}\Sigma_{\tilde{Q}} = \frac{-\Sigma_{\tilde{P}}}{6(\tilde{\pi}-\tau_-)} < 0.$$

In such case, Σ_P can not be any second-order statistics of distribution P from $\Sigma_P = E_P[\mathbf{x}\mathbf{x}^\top] \succeq 0$.

Hence, Lemma 2 presents the necessary yet insufficient conditions for some relations of data statistics in the noise and noise-free settings. To make noise estimation, we introduce two functions based on Lemma 2 as follows:

$$F(\tau) := \frac{\tilde{\pi}(1-\tau)}{\tilde{\pi}-\tau}\Sigma_{\tilde{P}} - \frac{\tau(1-\tilde{\pi})}{\tilde{\pi}-\tau}\Sigma_{\tilde{Q}} - \left[\frac{\tilde{\pi}(1-\tau)}{\tilde{\pi}-\tau}\mu_{\tilde{P}} - \frac{\tau(1-\tilde{\pi})}{\tilde{\pi}-\tau}\mu_{\tilde{Q}} \right] \left[\frac{\tilde{\pi}(1-\tau)}{\tilde{\pi}-\tau}\mu_{\tilde{P}} - \frac{\tau(1-\tilde{\pi})}{\tilde{\pi}-\tau}\mu_{\tilde{Q}} \right]^\top, \quad (8)$$

and

$$G(\tau) := \frac{(1-\tilde{\pi})(1-\tau)}{1-\tilde{\pi}-\tau}\Sigma_{\tilde{Q}} - \frac{\tilde{\pi}\tau}{1-\tilde{\pi}-\tau}\Sigma_{\tilde{P}} - \left[\frac{(1-\tilde{\pi})(1-\tau)}{1-\tilde{\pi}-\tau}\mu_{\tilde{Q}} - \frac{\tilde{\pi}\tau}{1-\tilde{\pi}-\tau}\mu_{\tilde{P}} \right] \left[\frac{(1-\tilde{\pi})(1-\tau)}{1-\tilde{\pi}-\tau}\mu_{\tilde{Q}} - \frac{\tilde{\pi}\tau}{1-\tilde{\pi}-\tau}\mu_{\tilde{P}} \right]^\top. \quad (9)$$

We have the following theorem for the positive semi-definiteness of $F(\tau)$ and $G(\tau)$, and the detailed proof is presented in Appendix C.

Theorem 1. If $\mu_{\tilde{P}}$, $\mu_{\tilde{Q}}$, $\Sigma_{\tilde{P}}$ and $\Sigma_{\tilde{Q}}$ are the first- and second-order statistics of distributions $\tilde{P}$ and $\tilde{Q}$, respectively, and $\tilde{\pi} = \Pr[\tilde{y} = +1]$, then there exists a unique $\tau_-^* \in [0, \tilde{\pi}]$ such that

$$F(\tau) \geq 0 \text{ for } \tau \in [0, \tau_-^*] \quad \text{and} \quad F(\tau) < 0 \text{ for } \tau \in (\tau_-^*, \tilde{\pi});$$

there also exists a unique $\tau_+^* \in [0, 1 - \tilde{\pi}]$ such that

$$G(\tau) \geq 0 \text{ for } \tau \in [0, \tau_+^*] \quad \text{and} \quad G(\tau) < 0 \text{ for } \tau \in (\tau_+^*, 1 - \tilde{\pi}).$$

Based on this theorem, we can estimate noise proportions by

$$\tau_-^* = \max \{ \tau \in [0, \tilde{\pi}] : F(\tau) \geq 0 \} \quad \text{and} \quad \tau_+^* = \max \{ \tau \in [0, 1 - \tilde{\pi}] : G(\tau) \geq 0 \}. \quad (10)$$

Theorem 1 guarantees the existence and uniqueness of noise estimation by Eqn. (10). As can be seen, this noise estimation does not require to estimate data distribution, as well as prior knowledge on data distribution. In addition, our noise estimation presents upper bounds on the ground-truth noise proportions τ_- and τ_+ as follows:

Corollary 1. Let τ_-^* and τ_+^* be the estimated noise proportions by Eqn. (10), and let τ_+ and τ_- be the ground-truth noise proportions. We have $\tau_- \leq \tau_-^*$ and $\tau_+ \leq \tau_+^*$.

This corollary follows directly from Theorem 1, Lemma 2 and the positive semi-definiteness of covariance matrix by Eqn. (10); in particular, we have

$$F(\tau_-) = \Sigma_P - \mu_P[\mu_P]^\top \succeq 0 \quad \text{and} \quad G(\tau_+) = \Sigma_Q - \mu_Q[\mu_Q]^\top \succeq 0,$$

where μ_P , μ_Q , Σ_P and Σ_Q are defined in Lemma 2.

Our noise estimation is partially motivated from [44], where Scott et al. [44] considered the largest noise proportions to ensure the existence of class distributions P and Q such that

$$\tilde{P} = \frac{\tilde{\pi} - \tau_-}{\tilde{\pi}(1 - \tau_-)}P + \frac{\tau_-(1 - \tilde{\pi})}{\tilde{\pi}(1 - \tau_-)}\tilde{Q} \quad \text{and} \quad \tilde{Q} = \frac{(1 - \tilde{\pi} - \tau_+)}{(1 - \tilde{\pi})(1 - \tau_+)}Q + \frac{\tau_+\tilde{\pi}}{(1 - \tilde{\pi})(1 - \tau_+)}\tilde{P}.$$

However, our work takes the largest noise proportions to guarantee the positive semi-definiteness of covariance matrix. Without prior knowledge of data distribution, previous studies on noise estimations [32,36,44], as well as our work, present upper bounds on the ground-truth noise proportions. This motivates us to further exploit the locality property of noise estimation, which will be shown in Section 6.

5. Convergence analysis

In the random noise model, we do not know the corrupted class distributions $\tilde{Q}$ and $\tilde{P}$, and what we can observe is a corrupted training sample $\tilde{S}_n = \{(\mathbf{x}_1, \tilde{y}_1), (\mathbf{x}_2, \tilde{y}_2), \dots, (\mathbf{x}_n, \tilde{y}_n)\}$. In this section, we consider noise estimation from the corrupted training sample $\tilde{S}_n$, and further present convergence analysis.

Let n_+ and n_- be the cardinalities of positive and negative instances in the corrupted training sample $\tilde{S}_n$, respectively, and write $\hat{\pi} = n_+/n$. We define the first- and second-order statistics of positive and negative instances in the corrupted training sample $\tilde{S}_n$, respectively, as

$$\hat{\mu}_{\tilde{P}} = \sum_{\tilde{y}_i=+1} \frac{\mathbf{x}_i}{n_+}, \quad \hat{\mu}_{\tilde{Q}} = \sum_{\tilde{y}_i=-1} \frac{\mathbf{x}_i}{n_-}, \quad \hat{\Sigma}_{\tilde{P}} = \sum_{\tilde{y}_i=+1} \frac{\mathbf{x}_i \mathbf{x}_i^\top}{n_+}, \quad \hat{\Sigma}_{\tilde{Q}} = \sum_{\tilde{y}_i=-1} \frac{\mathbf{x}_i \mathbf{x}_i^\top}{n_-}.$$

Similarly, we introduce two functions

$$\begin{aligned} \hat{F}(\tau) &:= \frac{\hat{\pi}(1-\tau)}{\hat{\pi}-\tau} \hat{\Sigma}_{\tilde{P}} - \frac{\tau(1-\hat{\pi})}{\hat{\pi}-\tau} \hat{\Sigma}_{\tilde{Q}} \\ &\quad - \left[\frac{\hat{\pi}(1-\tau)}{\hat{\pi}-\tau} \hat{\mu}_{\tilde{P}} - \frac{\tau(1-\hat{\pi})}{\hat{\pi}-\tau} \hat{\mu}_{\tilde{Q}} \right] \left[\frac{\hat{\pi}(1-\tau)}{\hat{\pi}-\tau} \hat{\mu}_{\tilde{P}} - \frac{\tau(1-\hat{\pi})}{\hat{\pi}-\tau} \hat{\mu}_{\tilde{Q}} \right]^\top, \end{aligned} \quad (11)$$

and

$$\begin{aligned} \hat{G}(\tau) &:= \frac{(1-\hat{\pi})(1-\tau)}{1-\hat{\pi}-\tau} \hat{\Sigma}_{\tilde{Q}} - \frac{\hat{\pi}\tau}{1-\hat{\pi}-\tau} \hat{\Sigma}_{\tilde{P}} \\ &\quad - \left[\frac{(1-\hat{\pi})(1-\tau)}{1-\hat{\pi}-\tau} \hat{\mu}_{\tilde{Q}} - \frac{\hat{\pi}\tau}{1-\hat{\pi}-\tau} \hat{\mu}_{\tilde{P}} \right] \left[\frac{(1-\hat{\pi})(1-\tau)}{1-\hat{\pi}-\tau} \hat{\mu}_{\tilde{Q}} - \frac{\tau\hat{\pi}}{1-\hat{\pi}-\tau} \hat{\mu}_{\tilde{P}} \right]^\top. \end{aligned} \quad (12)$$

Similarly to the proof of Theorem 1, we have

Theorem 2. If $\hat{\mu}_{\tilde{P}}$, $\hat{\mu}_{\tilde{Q}}$, $\hat{\Sigma}_{\tilde{P}}$ and $\hat{\Sigma}_{\tilde{Q}}$ are the first- and second-order statistics of positive and negative instances in the corrupted training sample $\tilde{S}_n$, respectively, and $\hat{\pi} = n_+/n$, then there exists a unique $\hat{\tau}_-^* \in [0, \hat{\pi}]$ such that

$$\hat{F}(\tau) \geq 0 \text{ for } \tau \in [0, \hat{\tau}_-^*] \quad \text{and} \quad \hat{F}(\tau) < 0 \text{ for } \tau \in (\hat{\tau}_-^*, \hat{\pi});$$

there also exists a unique $\hat{\tau}_+^* \in [0, 1 - \hat{\pi}]$ such that

$$\hat{G}(\tau) \geq 0 \text{ for } \tau \in [0, \hat{\tau}_+^*] \quad \text{and} \quad \hat{G}(\tau) < 0 \text{ for } \tau \in (\hat{\tau}_+^*, 1 - \hat{\pi}).$$

Based on this theorem, we can make noise estimation from the corrupted training sample $\tilde{S}_n$ by

$$\hat{\tau}_-^* = \max \left\{ \tau \in [0, \hat{\pi}]: \hat{F}(\tau) \geq 0 \right\} \quad \text{and} \quad \hat{\tau}_+^* = \max \left\{ \tau \in [0, 1 - \hat{\pi}]: \hat{G}(\tau) \geq 0 \right\}. \quad (13)$$

From Theorem 2, we also have

Corollary 2. Let $\hat{\tau}_-^*$ and $\hat{\tau}_+^*$ be the estimated noise proportions from Eqn. (13), and let τ_+ and τ_- be the ground-truth noise proportions. We have

$$\tau_- \leq \hat{\tau}_-^* \leq \hat{\pi} \quad \text{and} \quad \tau_+ \leq \hat{\tau}_+^* \leq 1 - \hat{\pi}$$

where $\hat{\pi}$ denotes the ratio of positive instances in the corrupted training sample $\tilde{S}_n$.

Recall that noise proportions τ_-^* and τ_+^* are estimated by Eqn. (10) from the corrupted class distributions $\tilde{P}$ and $\tilde{Q}$. We will present the convergence analysis on $|\hat{\tau}_-^* - \tau_-^*|$ in the subsequent of this section, i.e., the convergence of noise proportions estimated from the corrupted training sample and from the corrupted distribution. Similar analysis could be made to $|\hat{\tau}_+^* - \tau_+^*|$.

Recall $\tilde{\pi} = \Pr[\tilde{y} = +1]$, and we assume $\tilde{\pi}, \hat{\pi} \in (0, 1)$ without loss of generality. The covariance matrix of all instances in the corrupted training sample $\tilde{S}_n$ is given by

$$\widehat{\text{Cov}}_{D_{\mathcal{X}}} = \hat{\pi} \hat{\Sigma}_{\tilde{P}} + (1 - \hat{\pi}) \hat{\Sigma}_{\tilde{Q}} - \left[\hat{\pi} \hat{\mu}_{\tilde{P}} + (1 - \hat{\pi}) \hat{\mu}_{\tilde{Q}} \right] \left[\hat{\pi} \hat{\mu}_{\tilde{P}} + (1 - \hat{\pi}) \hat{\mu}_{\tilde{Q}} \right]^\top,$$

which can be viewed as an empirical estimation of

Algorithm 1 Binary search for the solution of $\hat{\tau}_-^*$.**Input:** The precision parameter σ , the ratio $\hat{\pi}$ of positive instances, and the first- and second-order statistics $\hat{\mu}_{\tilde{p}}, \hat{\mu}_{\tilde{q}}, \hat{\Sigma}_{\tilde{p}}$ and $\hat{\Sigma}_{\tilde{q}}$ **Initialize:** $l = 0$ and $r = \hat{\pi}$

```

1: while  $r - l \geq \sigma$  do
2:    $\tau = (l + r)/2$ 
3:   Calculate  $\hat{F}(\tau)$  according to Eqn. (11)
4:   if  $\hat{F}(\tau) \geq 0$  then
5:      $l = \tau$ 
6:   else
7:      $r = \tau$ 
8:   end if
9: end while

```

Output: $\hat{\tau}_-^* = (l + r)/2$

$$\text{Cov}_{D_{\mathcal{X}}} = \tilde{\pi} \Sigma_{\tilde{p}} + (1 - \tilde{\pi}) \Sigma_{\tilde{q}} - \left[\tilde{\pi} \mu_{\tilde{p}} + (1 - \tilde{\pi}) \mu_{\tilde{q}} \right] \left[\tilde{\pi} \mu_{\tilde{p}} + (1 - \tilde{\pi}) \mu_{\tilde{q}} \right]^T. \quad (14)$$

For technical analysis, we assume $\text{Cov}_{D_{\mathcal{X}}} \succ 0$ and $\widehat{\text{Cov}}_{D_{\mathcal{X}}} \succ 0$, which can be guaranteed by removing linear-dependent features. Based on those assumptions, we have

Theorem 3. For $\delta \in (0, 1)$ and instance space $\mathcal{X} = \{\mathbf{x} \in \mathbb{R}^d: \|\mathbf{x}\| \leq B\}$, the following holds with probability at least $1 - \delta$ over the corrupted training sample $\tilde{S}_n$:

$$|\hat{\tau}_-^* - \tau_-^*| \leq O\left(\sqrt[4]{(d \ln n_+ + \ln(1/\delta))/n_+}\right) + O\left(\sqrt[4]{(d \ln n_- + \ln(1/\delta))/n_-}\right);$$

the following also holds with probability at least $1 - \delta$ over the corrupted training sample $\tilde{S}_n$:

$$|\hat{\tau}_-^* - \tau_-^*| \leq O\left(\frac{1}{\rho} \sqrt{(d \ln n_+ + \ln(1/\delta))/n_+}\right) + O\left(\frac{1}{\rho} \sqrt{(d \ln n_- + \ln(1/\delta))/n_-}\right),$$

if there exists some $\rho \geq 0$ such that $\tau_-^* \leq \tilde{\pi} - \rho$ and $\hat{\tau}_-^* \leq \tilde{\pi} - \rho$.

This theorem shows the convergence between noise proportion $\hat{\tau}_-^*$ and τ_-^* , which are estimated from the corrupted training sample and corrupted data distribution, respectively. Theorem 3 does not require additional assumptions over data distribution, and the analysis is also independent of hypothesis space and kernel function. The detailed proof is given in Appendix D.

It is not easy to find some closed-form solutions for

$$\hat{\tau}_-^* = \max \left\{ \tau \in [0, \hat{\pi}]: \hat{F}(\tau) \geq 0 \right\} \quad \text{and} \quad \hat{\tau}_+^* = \max \left\{ \tau \in [0, 1 - \hat{\pi}]: \hat{G}(\tau) \geq 0 \right\},$$

and we turn to some numerical solutions. Based on Theorem 2, it is feasible to use the binary search to solve $\hat{\tau}_-^*$ and $\hat{\tau}_+^*$. Algorithm 1 presents the detailed procedure for the solution of $\hat{\tau}_-^*$, and we solve $\hat{\tau}_+^*$ in a similar manner.

6. The NESC method

Based on the noise estimation of Eqn. (13), it is not easy to make accurate estimation of the ground-truth noise proportions τ_- and τ_+ from the entire training sample $\tilde{S}_n$, which can be shown empirically in Table 4. This is because we merely provide upper bounds on the ground-truth noise proportions from Corollary 2, and statistics may be misled by random noise over the entire training sample, especially for some boundary instances.

In the random noise model, each observed label has been flipped independently with a certain probability, and this motivates us to exploit the locality property for noise estimation. In other words, we partition the instance space $\mathcal{X}$ into several continuous (maybe intersecting) areas $\mathcal{X}_1, \mathcal{X}_2, \dots, \mathcal{X}_k$, and make noise estimation in each area. The idea of locality has also been exploited to estimate class prior in positive and unlabeled data [3]. By using the locality of random noise and Corollary 2, we have

Proposition 1. Let $\mathcal{X}_1, \mathcal{X}_2, \dots, \mathcal{X}_k$ be continuous (maybe intersecting) areas in instance space $\mathcal{X}$. Let $\hat{\tau}_-^i$ and $\hat{\tau}_+^i$ be the noise proportions estimated from the training instances in area $\mathcal{X}_i$ by using Eqn. (13). We have

$$\tau_- \leq \min_{i \in [k]} \{\hat{\tau}_-^i\} \leq \min_{i \in [k]} \{\hat{\pi}_i\} \quad \text{and} \quad \tau_+ \leq \min_{i \in [k]} \{\hat{\tau}_+^i\} \leq \min_{i \in [k]} \{1 - \hat{\pi}_i\}$$

where τ_- and τ_+ are the ground-truth noise proportions, and $\hat{\pi}_i$ denotes the ratio of training positive instances in $\mathcal{X}_i$.

Without additional information on data distribution, this proposition still presents upper bounds for the ground-truth noise proportions even if we partition the instance space into several areas. We now discuss the estimation of ground-truth noise proportions. Liu and Tao [32] and Menon et al. [36] made noise estimation based on the assumption $\eta(\mathbf{x}) = 0$ and $\eta(\mathbf{x}') = 1$ for some $\mathbf{x}, \mathbf{x}' \in \mathcal{X}$, which is proven to be equivalent to the mutually irreducible assumption [44]. Motivated from [32,36], we could estimate the ground-truth noise proportions under some assumptions over data distribution by the following proposition, and the detailed proof is presented in Appendix E.

Proposition 2. Let $\eta(\mathbf{x})$ be the conditional probability of positive instances w.r.t. distribution D. For the ground-truth noise proportions τ_- and τ_+ , and for some continuous area $\mathcal{X}_1$, let $\hat{\tau}_-^*$ and $\hat{\tau}_+^*$ be the noise proportions estimated from the training instances in $\mathcal{X}_1$ by using Eqn. (13). We have $\hat{\tau}_-^* = \tau_-$ if $\eta(\mathbf{x}) = 0$ for all $\mathbf{x} \in \mathcal{X}_1$; and we also have $\hat{\tau}_+^* = \tau_+$ if $\eta(\mathbf{x}) = 1$ for all $\mathbf{x} \in \mathcal{X}_1$.

Proposition 2 presents a sufficient condition to estimate the ground-truth noise proportion. It is not easy to find some continuous area $\mathcal{X}_1$ in practice such that $\eta(\mathbf{x}) = 0$ (or $\eta(\mathbf{x}) = 1$) for each instance $\mathbf{x} \in \mathcal{X}_1$, whereas this motivates us to estimate the ground-truth noise proportions from some areas with relatively higher ratio of training positive/negative instances, as shown in Fig. 1. It is interesting to explore other assumptions over data distributions to estimate the ground-truth noise proportions, e.g., how difference between class distributions P and Q is sufficient for noise estimation from a theoretical view, and we leave it to future work.

Based on the previous analysis, we develop a simple and effective algorithm for noise estimation. The basic idea is to firstly use k -means algorithm to cluster the corrupted training data, and then make noise estimation for each cluster based on its first- and second-order statistics. We finally make noise estimation based on Proposition 1.

We now present the detailed estimation of τ_-^* in the subsequent of this section, and estimate τ_+^* similarly. Let $C_1^-, C_2^-, \dots, C_k^-$ denote k clusters by applying k -means algorithm to all negative instances in the corrupted sample $\tilde{S}_n$, where k is a hyper-parameter. For every $i \in [k]$, we calculate the center c_i^- and radius r_i^- of training negative instances in the cluster C_i^- as follows:

$$c_i^- = \sum_{j \in [n]: (\mathbf{x}_j, -1) \in C_i^- \cap \tilde{S}_n} \frac{\mathbf{x}_j}{|\{j' \in [n]: (\mathbf{x}_{j'}, -1) \in C_i^- \cap \tilde{S}_n\}|}, \quad (15)$$

$$r_i^- = \max \{ \|\mathbf{x}_j - c_i^-\| : j \in [n] \text{ and } (\mathbf{x}_j, -1) \in C_i^- \cap \tilde{S}_n \}, \quad (16)$$

and denote by

$$\mathcal{B}_i(r_i^-) = \{(\mathbf{x}_j, \tilde{y}_j) : (\mathbf{x}_j, \tilde{y}_j) \in \tilde{S}_n \text{ and } \|\mathbf{x}_j - c_i^-\| \leq r_i^-\}. \quad (17)$$

We make necessary adjustments on cluster C_i^- to ensure sufficient training data for the effective estimation of the first- and second-order statistics, and consider relatively higher ratio of negative instances for better noise estimation; therefore, we introduce an additional hyper-parameter $\beta > 0$, which is called size parameter, to adjust the cluster radius as follows:

- For small $\mathcal{B}_i(r_i^-)$ ($|\mathcal{B}_i(r_i^-)| < \beta$), it is insufficient to estimate the first- and second-order statistics without enough data; in such case, we increase the cluster radius

$$\tilde{r}_i^- = \min \{ r \geq r_i^- : |\mathcal{B}_i(r)| \geq \beta \}. \quad (18)$$

- For large $\mathcal{B}_i(r_i^-)$ ($|\mathcal{B}_i(r_i^-)| \geq \beta$), we update the radius with the highest ratio of negative instances in $\mathcal{B}_i(r_i^-)$ by

$$\tilde{r}_i^- = \arg \max_{r \leq r_i^- : |\mathcal{B}_i(r)| \geq \beta} \left\{ \left| \{(\mathbf{x}_i, \tilde{y}_i) \in \mathcal{B}_i(r) \cap \tilde{S}_n : \tilde{y}_i = -1\} \right| / |\mathcal{B}_i(r)| \right\}. \quad (19)$$

The intuitive explanation for the above is that, according to Propositions 1 and 2, it is relatively easy to estimate the ground-truth noise proportion τ_- from the data of high ratio of negative instances, under the premise of sufficient training data for statistics estimation.

Fixed the cluster radius $\tilde{r}_i^-$, we calculate the ratio $\hat{\pi}_i$ and statistics $\hat{\mu}_P^i$, $\hat{\mu}_Q^i$, $\hat{\Sigma}_P^i$ and $\hat{\Sigma}_Q^i$ of training instances in $\mathcal{B}_i(\tilde{r}_i^-)$, and estimate noise proportion by

$$\hat{\tau}_-^i = \max \left\{ \tau \in [0, \hat{\pi}_i] : \hat{F}_i(\tau) \geq 0 \right\},$$

where

$$\begin{aligned} \hat{F}_i(\tau) := & \frac{\hat{\pi}_i(1-\tau)}{\hat{\pi}_i-\tau} \hat{\Sigma}_P^i - \frac{\tau(1-\hat{\pi}_i)}{\hat{\pi}_i-\tau} \hat{\Sigma}_Q^i \\ & - \left[\frac{\hat{\pi}_i(1-\tau)}{\hat{\pi}_i-\tau} \hat{\mu}_P^i - \frac{\tau(1-\hat{\pi}_i)}{\hat{\pi}_i-\tau} \hat{\mu}_Q^i \right] \left[\frac{\hat{\pi}_i(1-\tau)}{\hat{\pi}_i-\tau} \hat{\mu}_P^i - \frac{\tau(1-\hat{\pi}_i)}{\hat{\pi}_i-\tau} \hat{\mu}_Q^i \right]^\top. \end{aligned} \quad (20)$$

Algorithm 2 Noise Estimation Statistics with Clusters (NESC) method.**Input:** Corrupted training sample $\tilde{S}_n$, k -means parameter k and size parameter β

```

1: Let  $C_1^-, C_2^-, \dots, C_k^-$  be  $k$  clusters by applying  $k$ -means algorithm to all negative instances in the corrupted training sample  $\tilde{S}_n$ 
2: for each cluster  $C_i^-$  ( $i \in [k]$ ) do
3:   Calculate the cluster  $c_i^-$ , radius  $r_i^-$ , and  $\mathcal{B}_i^-(r_i^-)$  according to Eqns. (15)–(17), respectively
4:   if  $|\mathcal{B}_i^-(r_i^-)| < \beta$  then
5:     Increase the cluster radius  $\tilde{r}_i^-$  according to Eqn. (18)
6:   else
7:     Update the cluster radius  $\tilde{r}_i^-$  according to Eqn. (19)
8:   end if
9:   Calculate the ratio  $\hat{\pi}_i$  and statistics  $\hat{\mu}_p^i, \hat{\mu}_Q^i, \hat{\Sigma}_p^i$  and  $\hat{\Sigma}_Q^i$  of training instances in  $\mathcal{B}_i^-(\tilde{r}_i^-)$ 
10:  Estimate  $\hat{\tau}_i^- \in [0, \hat{\pi}_i]$  by applying Algorithm 1 to Eqn. (20)
11: end for

```

Output: Noise proportion $\hat{\tau}_* = \min\{\hat{\tau}_1^-, \hat{\tau}_2^-, \dots, \hat{\tau}_k^-\}$ **Table 1**
Benchmark datasets (without noise corruption).

datasets	#instance	#feature	pos. ratio	datasets	#instance	#feature	pos. ratio
fourclass	862	2	0.3561	magic04	19,020	10	0.3516
optdigits	1,143	42	0.5004	shuttle	43,500	9	0.2159
gisette	6,000	5,000	0.5000	kddcup04	50,000	65	0.4972
landsat	6,435	36	0.2343	cod-rna	59,535	8	0.3333
letter	15,000	16	0.5061	sense	78,823	100	0.4997

We get the numerical solution of noise proportion $\hat{\tau}_i^-$ by Algorithm 1, and obtain the final noise proportion $\hat{\tau}_*$ from Proposition 1 by $\hat{\tau}_* = \min\{\hat{\tau}_1^-, \hat{\tau}_2^-, \dots, \hat{\tau}_k^-\}$. Algorithm 2 gives the detailed description of Noise Estimation Statistics with Clusters (NESC) method to estimate $\hat{\tau}_*$.

In comparison with previous work [32,36,41,44], our NESC is relatively easier to calculate the first- and second-order statistics, instead of learning class distributions, conditional distributions and kernel embedding of distributions. Our approach is independent of function space of data distribution and the choice of kernel functions, which may be easily implemented for real applications. In addition, our NESC method takes the least running-time costs, which will be shown empirically in Section 7.

7. Experiments

We evaluate our NESC method on ten benchmark datasets^{1,2}: fourclass, optdigits, gisette, landsat, letter, magic04, shuttle, kddcup04, cod-rna and sense, as summarized in Table 1. The features have been scaled to $[-1, 1]$ for all datasets, and multi-class datasets have been transformed into binary ones by randomly partitioning classes into two groups, where each group contains the same number of classes. We compare the NESC method with the state-of-the-art approaches on noise estimation as follows:

- **MPEMI**: Mixture proportion estimation based on mutual irreducibility [44];
- **KLIPE**³: Conditional probability for noise estimation [32];
- **MPEKED**⁴: Mixture proportion estimation via kernel embedding of distributions [41].

We consider the ground-truth noise proportions $(\tau_-, \tau_+) \in \{(0.05, 0.05), (0.2, 0.3), (0.4, 0.4)\}$. Given τ_- and τ_+ , labels of training data are flipped independently 30 times with different random seeds. For our NESC, we set $k = 4$ and $\beta = 200$ for fourclass with the smallest number of features, and set $k = 4$ and $\beta = 300$ for other datasets with features number smaller than 11; we also set $k = 10$ and $\beta = 1000$ for gisette, kddcup04 and sense with the largest number of features, and set $k = 6$ and $\beta = 600$ for other datasets.

For MPEMI, kernel density estimation is used for proportion estimation by Gaussian kernel with width 0.5. We take the default parameters for KLIPE as in [32]. There are two variants (KM1 and KM2) of kernel mean based gradient threshold algorithms in [41], while MPEKED only reports better noise estimation. We draw either 400, 800, 1600 and 3200 samples, and select Gaussian RBF kernels for KM1 and KM2 as in [41].

¹ <http://www.sigkdd.org/kddcup>, www.ics.uci.edu/~mllearn/MLRepository.html.

² We project the instances of datasets gisette and sense onto a 50-dimensional space by PCA as in [41].

³ The code is taken from <http://tongliangliu.esy.es>.

⁴ The code is taken from <http://web.eecs.umich.edu/~scott>.

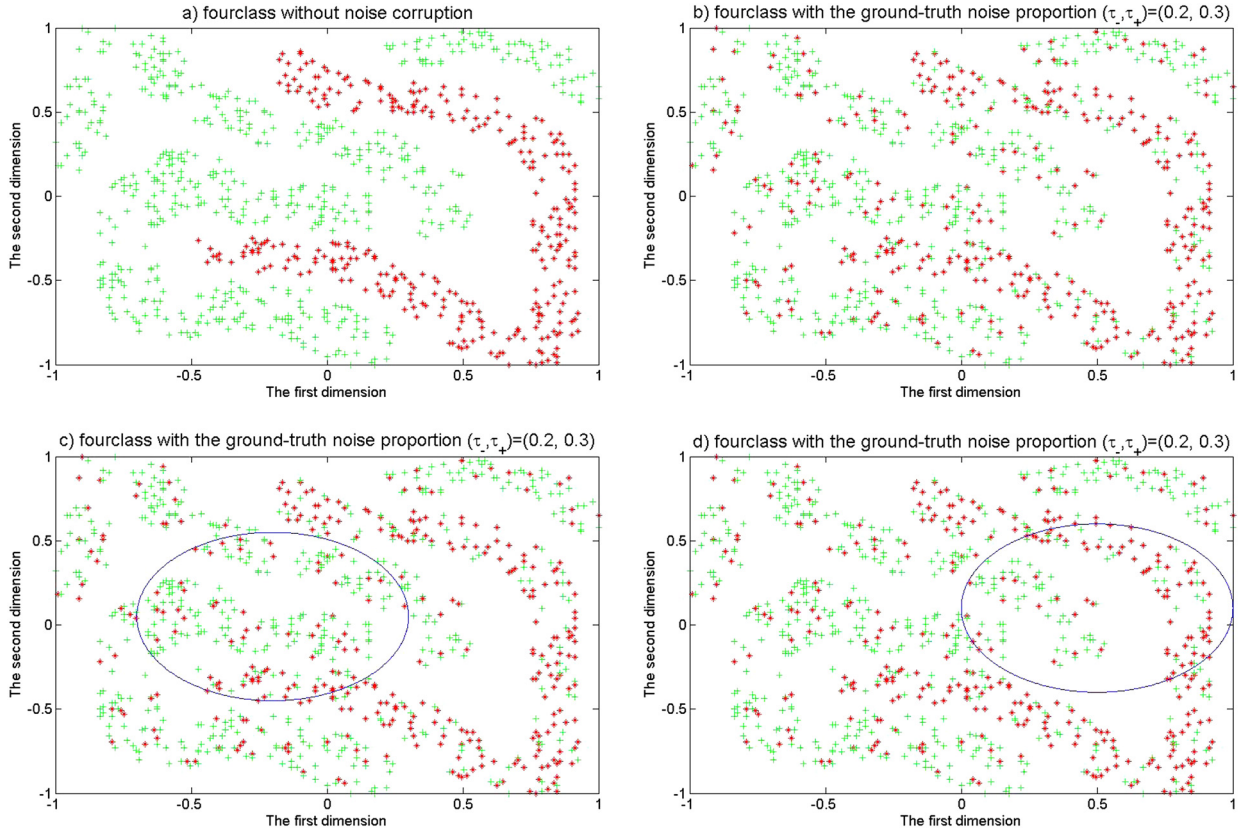


Fig. 1. Visualization of fourclass: a) clean dataset without noise corruption; b) corrupted dataset with the ground-truth noise proportion $(\tau_-, \tau_+) = (0.2, 0.3)$; c) corrupted dataset with $(\tau_-, \tau_+) = (0.2, 0.3)$ and our estimated noise proportion $(\hat{\tau}_-, \hat{\tau}_+) = (0.2298, 0.6914)$ from training instances in the circle; d) corrupted dataset with $(\tau_-, \tau_+) = (0.2, 0.3)$ and our estimated noise proportion $(\hat{\tau}_-, \hat{\tau}_+) = (0.3214, 0.2983)$ from training instances in the circle. Here, 'red star' and 'green plus' denote positive and negative instances, respectively. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

We first visualize dataset fourclass in Fig. 1 to show dataset with/without random noise, and noise estimation from training examples in different areas. Fig. 1-a) shows the clean data without noise corruption. Fig. 1-b) shows the corrupted dataset with the ground-truth noise proportion $(\tau_-, \tau_+) = (0.2, 0.3)$, and we estimate noise proportion $(\hat{\tau}_-, \hat{\tau}_+) = (0.2729, 0.4776)$ from the entire training instances by the NES method. Under the ground-truth noise proportion $(\tau_-, \tau_+) = (0.2, 0.3)$, our NES estimates noise proportion $(\hat{\tau}_-, \hat{\tau}_+) = (0.2298, 0.6914)$ from the training instances in the circle shown by Fig. 1-c), which implies that the estimated $\hat{\tau}_-$ is close to the ground-truth τ_- . Our NES also estimates noise proportion $(\hat{\tau}_-, \hat{\tau}_+) = (0.3214, 0.2983)$ from the training instances in the circle shown by Fig. 1-d), which implies that the estimated $\hat{\tau}_+$ is close to the ground-truth τ_+ . Therefore, our NES could estimate the ground-truth noise proportion from some areas with relatively high ratio of positive/negative instances.

Table 2 presents the empirical comparisons on noise estimation, which are calculated by averaging over 30 noise proportions with different random seeds. The performance is measured by the percentage of estimated noise proportions, whose deviations are smaller than 0.03 from the ground-truth noise proportions, i.e., $|\hat{\tau}_- - \tau_-| < 0.03$ and $|\hat{\tau}_+ - \tau_+| < 0.03$.

It is evident that our NES method achieves significantly better performance than other methods as shown in Table 2, since our noise estimations are closer to the ground-truth noise proportions in most times, e.g., NES takes 80% yet the others take less than 16% of the estimated noise proportions, whose deviations are smaller than 0.03 from the ground-truth noise proportions. This validates the effectiveness of our NES method empirically.

Tables 3 and 4 show the influence of clusters for the NES method, as well as the ratio of positive/negative instances in the 'winning' cluster, where the minimal noise proportion, i.e., $\min_{i \in [k]} \{\hat{\tau}_i^-\}$ or $\min_{i \in [k]} \{\hat{\tau}_i^+\}$, is estimated separately. We present the noise estimation of our NES method over the entire training data, and NES with 3 or 5 clusters and $\beta = 800$. Here, large size parameter $\beta = 800$ is set to guarantee sufficient statistical information of training instances in each cluster.

From Table 3, our NES method does not make accurate noise estimation over the entire training data, as it takes 25% of the estimated noise proportions, whose deviations are smaller than 0.03 from the ground-truth noise proportions. This is because an upper bound is provided over the ground-truth noise proportions from Corollary 2, and statistics may be misled by random noise, especially for some boundary instances. It is observable that, however, the noise estimation could be

Table 2

Comparisons of noise proportions (mean $\pm$ std.) of our NESC with other methods. We count the percentage of estimated noise proportions, whose deviations are smaller than 0.03 from the ground-truth noise proportions.

datasets	(τ_-, τ_+)	Our NESC	MPEMI	KLIPE	MPEKED
fourclass	(0.05, 0.05)	.0653 $\pm$.0070, .0781 $\pm$.0086	.1477 $\pm$.0037, .0734 $\pm$.0062	.0076 $\pm$.0607, .0682 $\pm$.0165	.1211 $\pm$.0401, .1403 $\pm$.0500
	(0.2, 0.3)	.1980 $\pm$.0169, .3046 $\pm$.0132	.1462 $\pm$.0073, .3130 $\pm$.0067	.1267 $\pm$.0218, .3079 $\pm$.0715	.2437 $\pm$.0615, .3607 $\pm$.0683
	(0.4, 0.4)	.3853 $\pm$.0275, .4158 $\pm$.0330	.4142 $\pm$.0092, .4396 $\pm$.0082	.4021 $\pm$.0462, .4724 $\pm$.0337	.4613 $\pm$.0837, .4683 $\pm$.0847
optdigits	(0.05, 0.05)	.0217 $\pm$.0059, .0284 $\pm$.0062	.1540 $\pm$.1067, .0003 $\pm$.0001	.0272 $\pm$.0030, .0167 $\pm$.0084	.0011 $\pm$.0008, .0023 $\pm$.0010
	(0.2, 0.3)	.1041 $\pm$.0203, .1512 $\pm$.0307	.1428 $\pm$.0489, .2653 $\pm$.0369	.1066 $\pm$.0136, .2327 $\pm$.0063	.1547 $\pm$.0418, .2039 $\pm$.0341
	(0.4, 0.4)	.4429 $\pm$.0104, .4367 $\pm$.0125	.3417 $\pm$.0434, .3625 $\pm$.0281	.3329 $\pm$.0266, .3415 $\pm$.0413	.3022 $\pm$.0367, .3243 $\pm$.0502
gisette	(0.05, 0.05)	.0290 $\pm$.0064, .0316 $\pm$.0054	.0000 $\pm$.0000, .0000 $\pm$.0000	.0000 $\pm$.0000, .0000 $\pm$.0000	.0000 $\pm$.0000, .0000 $\pm$.0000
	(0.2, 0.3)	.1460 $\pm$.0170, .2572 $\pm$.0147	.2660 $\pm$.0421, .2131 $\pm$.0391	.1308 $\pm$.0071, .2042 $\pm$.0090	.1410 $\pm$.0470, .2505 $\pm$.0382
	(0.4, 0.4)	.4267 $\pm$.0240, .4188 $\pm$.0152	.3071 $\pm$.0416, .3281 $\pm$.0281	.4547 $\pm$.0208, .3082 $\pm$.0271	.3463 $\pm$.0407, .3005 $\pm$.0642
landsat	(0.05, 0.05)	.0388 $\pm$.0099, .0632 $\pm$.0124	.9982 $\pm$.0032, .0000 $\pm$.0000	.5997 $\pm$.0181, .0001 $\pm$.0000	.2010 $\pm$.0123, .0004 $\pm$.0001
	(0.2, 0.3)	.1704 $\pm$.0097, .3198 $\pm$.0161	.2438 $\pm$.0312, .2617 $\pm$.0484	.1384 $\pm$.0136, .3718 $\pm$.0212	.2138 $\pm$.0682, .3279 $\pm$.0709
	(0.4, 0.4)	.3842 $\pm$.0148, .4020 $\pm$.0175	.4392 $\pm$.0503, .3647 $\pm$.0389	.3352 $\pm$.0199, .4782 $\pm$.0317	.3476 $\pm$.0814, .4610 $\pm$.0963
letter	(0.05, 0.05)	.0705 $\pm$.0095, .0371 $\pm$.0134	.0570 $\pm$.0067, .0976 $\pm$.0079	.0006 $\pm$.0024, .0170 $\pm$.0349	.0216 $\pm$.0051, .0343 $\pm$.0142
	(0.2, 0.3)	.1964 $\pm$.0465, .3258 $\pm$.0234	.1422 $\pm$.0503, .2594 $\pm$.0360	.1334 $\pm$.0286, .2666 $\pm$.0164	.2130 $\pm$.0497, .2607 $\pm$.0652
	(0.4, 0.4)	.4343 $\pm$.0384, .4286 $\pm$.0620	.3567 $\pm$.0805, .3613 $\pm$.0431	.3568 $\pm$.0832, .3492 $\pm$.0643	.3401 $\pm$.0715, .3293 $\pm$.1038
magic04	(0.05, 0.05)	.0695 $\pm$.0074, .0589 $\pm$.0066	.2697 $\pm$.0004, .3311 $\pm$.0027	.0208 $\pm$.0070, .0794 $\pm$.0022	.1612 $\pm$.0401, .2418 $\pm$.0714
	(0.2, 0.3)	.2127 $\pm$.0065, .3071 $\pm$.0068	.2426 $\pm$.0013, .3556 $\pm$.0083	.1086 $\pm$.0154, .2552 $\pm$.0258	.2305 $\pm$.0284, .2657 $\pm$.0527
	(0.4, 0.4)	.4002 $\pm$.0110, .4013 $\pm$.0108	.4489 $\pm$.0008, .4614 $\pm$.0043	.3268 $\pm$.0210, .3006 $\pm$.0084	.4736 $\pm$.0741, .3314 $\pm$.0986
shuttle	(0.05, 0.05)	.0484 $\pm$.0017, .0457 $\pm$.0062	.1207 $\pm$.0029, .0936 $\pm$.0120	.0001 $\pm$.0001, .0001 $\pm$.0001	.1823 $\pm$.0379, .1614 $\pm$.0257
	(0.2, 0.3)	.1789 $\pm$.0256, .2702 $\pm$.0423	.1892 $\pm$.0041, .2614 $\pm$.0148	.1403 $\pm$.0512, .2320 $\pm$.0571	.2509 $\pm$.0745, .2829 $\pm$.0401
	(0.4, 0.4)	.3939 $\pm$.0083, .3903 $\pm$.0063	.3908 $\pm$.0796, .3487 $\pm$.0956	.2701 $\pm$.0228, .2908 $\pm$.0061	.4865 $\pm$.0614, .4792 $\pm$.0683
kddcup04	(0.05, 0.05)	.0323 $\pm$.0409, .0611 $\pm$.0354	.0002 $\pm$.0001, .0001 $\pm$.0001	.3220 $\pm$.0033, .3274 $\pm$.0036	.0000 $\pm$.0000, .0000 $\pm$.0000
	(0.2, 0.3)	.2692 $\pm$.0108, .3580 $\pm$.0137	.1760 $\pm$.0486, .2317 $\pm$.0193	.2447 $\pm$.0253, .4074 $\pm$.0473	.0971 $\pm$.0767, .2198 $\pm$.0623
	(0.4, 0.4)	.4230 $\pm$.0125, .4492 $\pm$.0135	.4635 $\pm$.0574, .4307 $\pm$.0492	.4577 $\pm$.0353, .4588 $\pm$.0409	.3027 $\pm$.0695, .3188 $\pm$.0857
cod-rna	(0.05, 0.05)	.0645 $\pm$.0031, .0896 $\pm$.0032	.1302 $\pm$.0302, .0852 $\pm$.0065	.0002 $\pm$.0001, .1201 $\pm$.0452	.0018 $\pm$.0080, .1635 $\pm$.0487
	(0.2, 0.3)	.2193 $\pm$.0082, .3295 $\pm$.0070	.1407 $\pm$.0426, .2074 $\pm$.0568	.1615 $\pm$.0327, .3462 $\pm$.0429	.1240 $\pm$.0352, .2148 $\pm$.0419
	(0.4, 0.4)	.4036 $\pm$.0051, .4164 $\pm$.0052	.4328 $\pm$.0439, .3265 $\pm$.0485	.3427 $\pm$.0507, .4619 $\pm$.0484	.3085 $\pm$.0413, .3405 $\pm$.0528
sense	(0.05, 0.05)	.1228 $\pm$.0140, .0556 $\pm$.0071	.0000 $\pm$.0000, .0000 $\pm$.0000	.0285 $\pm$.0003, .0321 $\pm$.0049	.0001 $\pm$.0001, .0003 $\pm$.0000
	(0.2, 0.3)	.2245 $\pm$.0270, .3017 $\pm$.0087	.2491 $\pm$.0294, .2282 $\pm$.0517	.1387 $\pm$.0056, .2319 $\pm$.0020	.1679 $\pm$.0269, .2187 $\pm$.0351
	(0.4, 0.4)	.4068 $\pm$.0051, .3912 $\pm$.0062	.4436 $\pm$.0360, .3263 $\pm$.0539	.3349 $\pm$.0038, .3322 $\pm$.0012	.3184 $\pm$.0419, .3095 $\pm$.0585
Deviation < 0.03		80.00%	13.33%	15.00%	10.00%

improved by considering proper clusters for the NESC method; for example, we achieve 46.67% and 56.67% of the estimated noise proportions by NESC with 3 and 5 clusters, respectively, as shown in Table 4.

For small random noise proportions $(\tau_-, \tau_+) \in \{(0.05, 0.05), (0.2, 0.3)\}$, our NESC method could make good noise estimation according to the first- and second-order statistics of the corrupted data, and the estimated noise proportions are far from the ratios of positive/negative instances in the ‘winning’ cluster, as shown in Tables 3 and 4. It is also observable, however, that clusters are helpful to estimate large noise proportions such as $(\tau_-, \tau_+) = (0.4, 0.4)$, where random noise almost dominates the labels of training instances, and the corrupted class distributions $\tilde{P}$ and $\tilde{Q}$ are close each other. In such case, the noise estimation by NESC depends on the ratios of training positive/negative instances for clusters, which have been close to the ground-truth noise proportions in many times.

We also compare the average CPU time cost of NESC with other three methods. As can be seen from Fig. 2, our NESC takes the least running time, particularly for larger datasets, where NESC is about 100 times faster than KLIPE and MPEKED, and 10 times than MPEMI. This is because NESC takes almost linear time complexity when the iteration number and parameter k are small for k -means algorithm, whereas the others require quadratic or cubic time complexity.

8. Conclusion

This work focuses on a fundamental problem: how to make noise estimation from corrupted data. We develop a simple and effective method for noise estimation, and the basic idea is to utilize the first- and second-order statistics of observed data, as well as the positive semi-definiteness of covariance matrices. Then, an upper bound on noise estimation is provided without additional assumptions over data distribution. We develop the Noise Estimation Statistics with Clusters (NESC) method for noise estimation. Theoretically, we present the existence, uniqueness and convergence analysis of our noise

Table 3

Noise proportions (mean $\pm$ std.) of our NESC method over the entire training data, as well as clusters used in Table 2. We also show the ratio of training positive/negative instances in the 'winning' cluster by NESC, in which the minimal noise proportion is estimated separately.

datasets	(τ_- , τ_+)	NESC on the entire training data	pos./neg. ratio of the entire training data	NESC in Table 2	pos./neg. ratio of the 'winning' cluster by NESC (Table 2)
fourclass	(0.05, 0.05)	.1623 $\pm$.0027, .3481 $\pm$.0035	.3779 $\pm$.0018, .6221 $\pm$.0018	.0653 $\pm$.0070, .0781 $\pm$.0086	.1917 $\pm$.0841, .2753 $\pm$.0484
	(0.2, 0.3)	.2694 $\pm$.0030, .4570 $\pm$.0039	.3875 $\pm$.0028, .6125 $\pm$.0028	.1980 $\pm$.0169, .3046 $\pm$.0132	.3061 $\pm$.0526, .4119 $\pm$.0293
	(0.4, 0.4)	.4222 $\pm$.0058, .4669 $\pm$.0056	.4725 $\pm$.0025, .5275 $\pm$.0025	.3853 $\pm$.0275, .4158 $\pm$.0330	.4305 $\pm$.0259, .4606 $\pm$.0268
optdigits	(0.05, 0.05)	.0440 $\pm$.0073, .0449 $\pm$.0042	.5004 $\pm$.0011, .4996 $\pm$.0011	.0217 $\pm$.0059, .0284 $\pm$.0062	.3644 $\pm$.0413, .2609 $\pm$.0269
	(0.2, 0.3)	.2423 $\pm$.0131, .3221 $\pm$.0140	.4499 $\pm$.0009, .5501 $\pm$.0009	.1041 $\pm$.0203, .1512 $\pm$.0307	.3849 $\pm$.0465, .4594 $\pm$.0453
	(0.4, 0.4)	.4926 $\pm$.0024, .4853 $\pm$.0026	.5000 $\pm$.0023, .5000 $\pm$.0023	.4429 $\pm$.0104, .4367 $\pm$.0125	.4567 $\pm$.0107, .4423 $\pm$.0130
gisette	(0.05, 0.05)	.0767 $\pm$.0052, .0903 $\pm$.0037	.5000 $\pm$.0002, .5000 $\pm$.0002	.0290 $\pm$.0064, .0316 $\pm$.0054	.1642 $\pm$.0779, .1717 $\pm$.0391
	(0.2, 0.3)	.2069 $\pm$.0063, .3110 $\pm$.0090	.4499 $\pm$.0001, .5501 $\pm$.0001	.1460 $\pm$.0170, .2572 $\pm$.0147	.3113 $\pm$.0486, .4617 $\pm$.0372
	(0.4, 0.4)	.4816 $\pm$.0005, .4989 $\pm$.0003	.5000 $\pm$.0003, .5000 $\pm$.0003	.4267 $\pm$.0240, .4188 $\pm$.0152	.4544 $\pm$.0329, .4471 $\pm$.0263
landsat	(0.05, 0.05)	.0452 $\pm$.0047, .4445 $\pm$.0024	.2609 $\pm$.0002, .7391 $\pm$.0002	.0388 $\pm$.0099, .0632 $\pm$.0124	.0627 $\pm$.0386, .0660 $\pm$.0167
	(0.2, 0.3)	.1684 $\pm$.0128, .5145 $\pm$.0043	.3171 $\pm$.0001, .6829 $\pm$.0001	.1704 $\pm$.0097, .3198 $\pm$.0161	.3354 $\pm$.0881, .3198 $\pm$.0161
	(0.4, 0.4)	.4466 $\pm$.0002, .5521 $\pm$.0002	.4469 $\pm$.0002, .5531 $\pm$.0002	.3842 $\pm$.0148, .4020 $\pm$.0175	.3940 $\pm$.0217, .4020 $\pm$.0175
letter	(0.05, 0.05)	.1590 $\pm$.0034, .1493 $\pm$.0024	.5048 $\pm$.0005, .4952 $\pm$.0005	.0705 $\pm$.0095, .0371 $\pm$.0134	.2912 $\pm$.0648, .1721 $\pm$.0782
	(0.2, 0.3)	.2561 $\pm$.0057, .3556 $\pm$.0036	.4492 $\pm$.0006, .5508 $\pm$.0006	.1964 $\pm$.0465, .3258 $\pm$.0234	.3830 $\pm$.0735, .4886 $\pm$.0683
	(0.4, 0.4)	.5013 $\pm$.0001, .4916 $\pm$.0001	.5013 $\pm$.0001, .4987 $\pm$.0001	.4343 $\pm$.0384, .4286 $\pm$.0620	.4512 $\pm$.0564, .4337 $\pm$.0633
magic04	(0.05, 0.05)	.1891 $\pm$.0008, .0994 $\pm$.0055	.3664 $\pm$.0002, .6336 $\pm$.0002	.0695 $\pm$.0074, .0589 $\pm$.0066	.1204 $\pm$.0354, .2034 $\pm$.0502
	(0.2, 0.3)	.2770 $\pm$.0012, .3147 $\pm$.0146	.3755 $\pm$.0002, .6245 $\pm$.0002	.2127 $\pm$.0065, .3071 $\pm$.0068	.2645 $\pm$.0317, .3027 $\pm$.0481
	(0.4, 0.4)	.4703 $\pm$.0002, .5297 $\pm$.0002	.4703 $\pm$.0002, .5297 $\pm$.0002	.4002 $\pm$.0110, .4013 $\pm$.0108	.4057 $\pm$.0175, .4071 $\pm$.0204
shuttle	(0.05, 0.05)	.1082 $\pm$.0003, .0901 $\pm$.0009	.2445 $\pm$.0001, .7555 $\pm$.0001	.0484 $\pm$.0017, .0457 $\pm$.0062	.0934 $\pm$.0383, .0557 $\pm$.0126
	(0.2, 0.3)	.1999 $\pm$.0419, .3015 $\pm$.0378	.3080 $\pm$.0001, .6920 $\pm$.0001	.1789 $\pm$.0256, .2702 $\pm$.0423	.2509 $\pm$.0688, .4629 $\pm$.1063
	(0.4, 0.4)	.4412 $\pm$.0003, .5567 $\pm$.0001	.4433 $\pm$.0001, .5567 $\pm$.0001	.3939 $\pm$.0083, .3903 $\pm$.0063	.3979 $\pm$.0164, .4213 $\pm$.0099
kddcup04	(0.05, 0.05)	.1349 $\pm$.0057, .1280 $\pm$.0036	.4975 $\pm$.0001, .5025 $\pm$.0001	.0323 $\pm$.0409, .0611 $\pm$.0354	.4175 $\pm$.1740, .4554 $\pm$.1529
	(0.2, 0.3)	.4480 $\pm$.0003, .5015 $\pm$.0006	.4485 $\pm$.0001, .5515 $\pm$.0001	.2692 $\pm$.0108, .3580 $\pm$.0137	.3018 $\pm$.0125, .4369 $\pm$.0155
	(0.4, 0.4)	.4985 $\pm$.0002, .4902 $\pm$.0008	.4995 $\pm$.0000, .5005 $\pm$.0000	.4230 $\pm$.0125, .4492 $\pm$.0135	.4230 $\pm$.0125, .4492 $\pm$.0135
cod-rna	(0.05, 0.05)	.0886 $\pm$.0005, .1846 $\pm$.0002	.3500 $\pm$.0000, .6500 $\pm$.0000	.0645 $\pm$.0031, .0896 $\pm$.0032	.2323 $\pm$.0072, .1227 $\pm$.0042
	(0.2, 0.3)	.2212 $\pm$.0009, .3752 $\pm$.0007	.3666 $\pm$.0001, .6334 $\pm$.0001	.2193 $\pm$.0082, .3295 $\pm$.0070	.2193 $\pm$.0082, .3295 $\pm$.0070
	(0.4, 0.4)	.4667 $\pm$.0005, .5301 $\pm$.0052	.4667 $\pm$.0005, .5333 $\pm$.0005	.4036 $\pm$.0051, .4164 $\pm$.0052	.4036 $\pm$.0051, .4164 $\pm$.0052
sense	(0.05, 0.05)	.1352 $\pm$.0032, .1185 $\pm$.0005	.4997 $\pm$.0001, .5003 $\pm$.0001	.1228 $\pm$.0140, .0556 $\pm$.0071	.2908 $\pm$.1283, .1056 $\pm$.0271
	(0.2, 0.3)	.2134 $\pm$.0152, .2998 $\pm$.0099	.4498 $\pm$.0001, .5502 $\pm$.0001	.2245 $\pm$.0270, .3017 $\pm$.0087	.2445 $\pm$.0069, .3017 $\pm$.0058
	(0.4, 0.4)	.4400 $\pm$.0002, .4202 $\pm$.0001	.4999 $\pm$.0000, .5001 $\pm$.0000	.4068 $\pm$.0051, .3912 $\pm$.0062	.4186 $\pm$.0085, .3912 $\pm$.0062
Deviation < 0.03		25.00%	0.00%	80.00%	31.67%

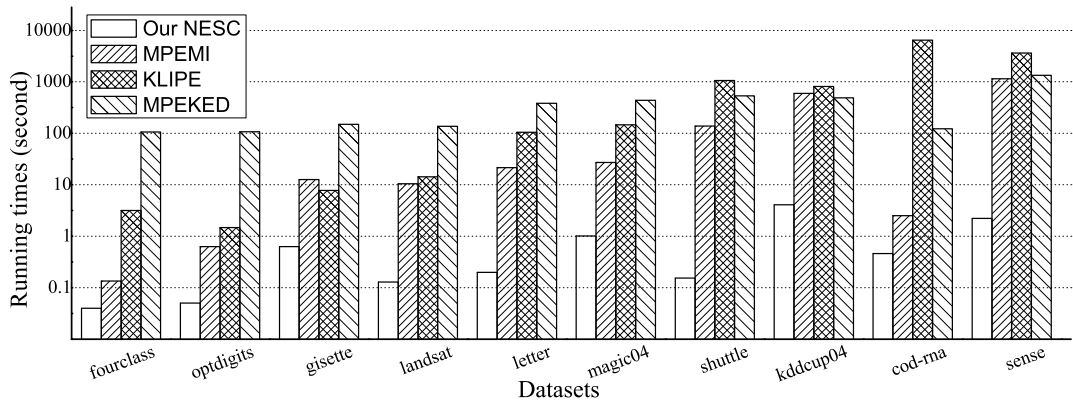
**Fig. 2.** Comparisons of the running time (in seconds) of our NESC and other three methods. Notice that the y-axis is in log-scale.

Table 4

Noise proportions (mean $\pm$ std.) of our NESC method with 3 or 5 clusters and $\beta = 800$. We also show the ratio of training positive/negative instances in the 'winning' cluster by NESC, in which the minimal noise proportion is estimated separately.

datasets	(τ_- , τ_+)	NESC (3 clusters)	pos./neg. ratio of the 'winning' cluster by NESC (3 clusters)	NESC (5 clusters)	pos./neg. ratio of the 'winning' cluster by NESC (5 clusters)
fourclass	(0.05, 0.05)	.1579 $\pm$.0029, .2856 $\pm$.0096	.3429 $\pm$.0047, .6015 $\pm$.0029	.1580 $\pm$.0027, .2829 $\pm$.0084	.3414 $\pm$.0056, .6014 $\pm$.0027
	(0.2, 0.3)	.2666 $\pm$.0030, .4270 $\pm$.0065	.3718 $\pm$.0111, .6011 $\pm$.0032	.2665 $\pm$.0031, .4195 $\pm$.0059	.3717 $\pm$.0093, .5993 $\pm$.0030
	(0.4, 0.4)	.4191 $\pm$.0070, .4521 $\pm$.0080	.4682 $\pm$.0056, .5223 $\pm$.0049	.4179 $\pm$.0065, .4484 $\pm$.0065	.4698 $\pm$.0062, .5221 $\pm$.0032
optdigits	(0.05, 0.05)	.0390 $\pm$.0067, .0322 $\pm$.0030	.4676 $\pm$.0410, .3968 $\pm$.0354	.0355 $\pm$.0079, .0362 $\pm$.0026	.4710 $\pm$.0385, .3804 $\pm$.0440
	(0.2, 0.3)	.1623 $\pm$.0208, .1858 $\pm$.0166	.4316 $\pm$.0192, .4996 $\pm$.0355	.1525 $\pm$.0178, .1675 $\pm$.0143	.4316 $\pm$.0236, .4979 $\pm$.0348
	(0.4, 0.4)	.4840 $\pm$.0066, .4751 $\pm$.0069	.4970 $\pm$.0057, .4858 $\pm$.0062	.4753 $\pm$.0071, .4615 $\pm$.0073	.5038 $\pm$.0071, .4936 $\pm$.0054
gisette	(0.05, 0.05)	.0389 $\pm$.0063, .0399 $\pm$.0055	.1381 $\pm$.0772, .1720 $\pm$.0390	.0305 $\pm$.0073, .0321 $\pm$.0064	.1335 $\pm$.0328, .1393 $\pm$.0268
	(0.2, 0.3)	.2264 $\pm$.0269, .3047 $\pm$.0380	.3749 $\pm$.0442, .5517 $\pm$.0755	.1926 $\pm$.0229, .2889 $\pm$.0437	.2923 $\pm$.0418, .4891 $\pm$.0524
	(0.4, 0.4)	.4153 $\pm$.0127, .4411 $\pm$.0091	.4226 $\pm$.0151, .4424 $\pm$.0101	.4193 $\pm$.0119, .4161 $\pm$.0123	.4223 $\pm$.0133, .4174 $\pm$.0138
landsat	(0.05, 0.05)	.0465 $\pm$.0037, .1017 $\pm$.0107	.1967 $\pm$.1029, .1258 $\pm$.0829	.0435 $\pm$.0045, .0859 $\pm$.0076	.1202 $\pm$.0809, .0956 $\pm$.0359
	(0.2, 0.3)	.1664 $\pm$.0159, .3480 $\pm$.0069	.3414 $\pm$.0372, .3591 $\pm$.0172	.1633 $\pm$.0152, .3316 $\pm$.0095	.3523 $\pm$.0479, .3420 $\pm$.0158
	(0.4, 0.4)	.3960 $\pm$.0067, .4359 $\pm$.0286	.4055 $\pm$.0073, .4550 $\pm$.0461	.3901 $\pm$.0061, .4179 $\pm$.0105	.3998 $\pm$.0058, .4370 $\pm$.0106
letter	(0.05, 0.05)	.1053 $\pm$.0262, .0607 $\pm$.0103	.4080 $\pm$.0962, .2409 $\pm$.0425	.0722 $\pm$.0127, .0497 $\pm$.0113	.3315 $\pm$.0677, .2035 $\pm$.0603
	(0.2, 0.3)	.2612 $\pm$.0348, .3540 $\pm$.0085	.4304 $\pm$.0334, .5181 $\pm$.0607	.2347 $\pm$.0431, .3181 $\pm$.0286	.3969 $\pm$.0445, .5258 $\pm$.0381
	(0.4, 0.4)	.4846 $\pm$.0099, .4481 $\pm$.0153	.4872 $\pm$.0167, .4522 $\pm$.0168	.4703 $\pm$.0090, .4373 $\pm$.0132	.4731 $\pm$.0117, .4394 $\pm$.0149
magic04	(0.05, 0.05)	.1099 $\pm$.0128, .0727 $\pm$.0405	.4401 $\pm$.0813, .4457 $\pm$.0786	.0933 $\pm$.0065, .0698 $\pm$.0189	.4951 $\pm$.0632, .4796 $\pm$.0581
	(0.2, 0.3)	.2445 $\pm$.0051, .3273 $\pm$.0114	.3560 $\pm$.0233, .4618 $\pm$.0207	.2218 $\pm$.0071, .3116 $\pm$.0120	.3264 $\pm$.0117, .4395 $\pm$.0162
	(0.4, 0.4)	.4173 $\pm$.0039, .4320 $\pm$.0102	.4658 $\pm$.0640, .4695 $\pm$.0746	.4049 $\pm$.0093, .4109 $\pm$.0076	.4546 $\pm$.0098, .4528 $\pm$.0064
shuttle	(0.05, 0.05)	.0485 $\pm$.0019, .0472 $\pm$.0017	.0907 $\pm$.0507, .0547 $\pm$.0015	.0482 $\pm$.0020, .0464 $\pm$.0016	.0807 $\pm$.0312, .0536 $\pm$.0076
	(0.2, 0.3)	.1893 $\pm$.0150, .2798 $\pm$.0302	.2531 $\pm$.0939, .4871 $\pm$.1279	.1757 $\pm$.0324, .2640 $\pm$.0427	.2544 $\pm$.0900, .4816 $\pm$.1521
	(0.4, 0.4)	.3968 $\pm$.0034, .3943 $\pm$.0046	.4149 $\pm$.0144, .4346 $\pm$.0146	.3936 $\pm$.0076, .3911 $\pm$.0116	.4047 $\pm$.0093, .4173 $\pm$.0075
kddcup04	(0.05, 0.05)	.1530 $\pm$.0611, .1468 $\pm$.0669	.4816 $\pm$.0516, .4857 $\pm$.0493	.0975 $\pm$.0430, .0867 $\pm$.0394	.5076 $\pm$.0838, .4767 $\pm$.0656
	(0.2, 0.3)	.3472 $\pm$.0451, .4324 $\pm$.0238	.3662 $\pm$.0527, .4596 $\pm$.0600	.3052 $\pm$.0074, .3876 $\pm$.0136	.3333 $\pm$.0131, .4483 $\pm$.0209
	(0.4, 0.4)	.4741 $\pm$.0149, .4757 $\pm$.0131	.4856 $\pm$.0153, .4797 $\pm$.0139	.4578 $\pm$.0108, .4318 $\pm$.0055	.4644 $\pm$.0098, .4528 $\pm$.0115
cod-rna	(0.05, 0.05)	.0659 $\pm$.0006, .1239 $\pm$.0011	.2578 $\pm$.0003, .1239 $\pm$.0011	.0640 $\pm$.0006, .1023 $\pm$.0022	.2454 $\pm$.0008, .1119 $\pm$.0021
	(0.2, 0.3)	.2089 $\pm$.0056, .3443 $\pm$.0019	.2188 $\pm$.0058, .3473 $\pm$.0020	.2109 $\pm$.0041, .3205 $\pm$.0054	.2178 $\pm$.0051, .3263 $\pm$.0079
	(0.4, 0.4)	.4072 $\pm$.0036, .4178 $\pm$.0023	.4097 $\pm$.0043, .4278 $\pm$.0072	.4018 $\pm$.0026, .4111 $\pm$.0035	.4032 $\pm$.0036, .4212 $\pm$.0038
sense	(0.05, 0.05)	.1340 $\pm$.0028, .0660 $\pm$.0075	.4318 $\pm$.0836, .1643 $\pm$.0471	.1335 $\pm$.0031, .0571 $\pm$.0036	.3673 $\pm$.0920, .0956 $\pm$.0236
	(0.2, 0.3)	.2504 $\pm$.0084, .3083 $\pm$.0084	.3032 $\pm$.0074, .3474 $\pm$.0087	.2481 $\pm$.0078, .3046 $\pm$.0091	.2589 $\pm$.0061, .3093 $\pm$.0079
	(0.4, 0.4)	.4156 $\pm$.0110, .4019 $\pm$.0066	.4249 $\pm$.0117, .4107 $\pm$.0073	.4084 $\pm$.0098, .3990 $\pm$.0083	.4110 $\pm$.0096, .3990 $\pm$.0083
Deviation < 0.03		46.67%	15.00%	56.67%	21.67%

estimation, and empirical studies show that our NESC method achieves significantly better performance than state-of-the-art approaches on noise estimation, also with the lowest running-time cost.

This work is part of the story for classification in the presence of label noise. An interesting future work is to develop some effective algorithms such as kernel SVMs [48] to deal with noisy labels based on our noise estimation, and analyze the influence of inaccurate noise estimation and performance obtained from some specific learning algorithms. Another interesting future work is to generalize our idea of this work to the noise estimation of multi-class learning, where one challenge lies in the accurate estimation of statistics, especially for the class-imbalance tasks with a large amount of classes and scarce training instances for minority classes. It is also interesting to study noise estimations for some new paradigms such as abductive learning [49], which encompasses machine learning and logical reasoning.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Proof of Lemma 1

We first give the ratio of positive instances w.r.t. ground-truth distribution D as

$$\pi = \Pr[y = +1] = \int_{\mathbf{x} \in \mathcal{X}} \eta(\mathbf{x}) f_{D_{\mathcal{X}}}(\mathbf{x}) d\mathbf{x}.$$

Based on the Bayes' theorem, we have

$$f_P(\mathbf{x}) = \Pr[\mathbf{x}|y = +1] = \Pr[\mathbf{x}] \Pr[y = +1|\mathbf{x}] / \Pr[y = +1] = \eta(\mathbf{x}) f_{D_{\mathcal{X}}}(\mathbf{x}) / \pi. \quad (21)$$

Similarly, we have

$$f_Q(\mathbf{x}) = (1 - \eta(\mathbf{x})) f_{D_{\mathcal{X}}}(\mathbf{x}) / (1 - \pi).$$

For the corrupted distribution $\tilde{D}$, we have

$$\tilde{\eta}(\mathbf{x}) = (1 - \tau_+) \eta(\mathbf{x}) + \tau_-(1 - \eta(\mathbf{x})) = (1 - \tau_+ - \tau_-) \eta(\mathbf{x}) + \tau_-, \quad (22)$$

which yields that

$$\tilde{\pi} = \int_{\mathbf{x} \in \mathcal{X}} \tilde{\eta}(\mathbf{x}) f_{D_{\mathcal{X}}}(\mathbf{x}) d\mathbf{x} = (1 - \tau_+ - \tau_-) \pi + \tau_-. \quad (23)$$

This proves $\tau_- \leq \tilde{\pi}$ from $\tau_+ + \tau_- \leq 1$ and $\pi \in [0, 1]$. From Eqn. (23), we also have

$$1 - \tilde{\pi} = (1 - \tau_+ - \tau_-)(1 - \pi) + \tau_+,$$

which proves $\tau_+ \leq 1 - \tilde{\pi}$. This follows that

$$\begin{bmatrix} \tilde{\pi} \\ 1 - \tilde{\pi} \end{bmatrix} = \begin{bmatrix} 1 - \tau_+ & \tau_- \\ \tau_+ & 1 - \tau_- \end{bmatrix} \begin{bmatrix} \pi \\ 1 - \pi \end{bmatrix}.$$

In a similar manner, we have

$$f_{\tilde{P}}(\mathbf{x}) = \tilde{\eta}(\mathbf{x}) f_{D_{\mathcal{X}}}(\mathbf{x}) / \tilde{\pi} \quad \text{and} \quad f_{\tilde{Q}}(\mathbf{x}) = (1 - \tilde{\eta}(\mathbf{x})) f_{D_{\mathcal{X}}}(\mathbf{x}) / (1 - \tilde{\pi})$$

which proves

$$f_{D_{\mathcal{X}}}(\mathbf{x}) = \tilde{\pi} f_{\tilde{P}}(\mathbf{x}) + (1 - \tilde{\pi}) f_{\tilde{Q}}(\mathbf{x}).$$

From Eqns. (21)-(23), we further have

$$\tilde{\pi} f_{\tilde{P}}(\mathbf{x}) = \tilde{\eta}(\mathbf{x}) f_{D_{\mathcal{X}}}(\mathbf{x}) = (1 - \tau_- - \tau_+) \eta(\mathbf{x}) f_{D_{\mathcal{X}}}(\mathbf{x}) + \tau_- f_{D_{\mathcal{X}}}(\mathbf{x}) = (\tilde{\pi} - \tau_-) f_P(\mathbf{x}) + \tau_- f_{D_{\mathcal{X}}}(\mathbf{x}),$$

which yields that

$$\tilde{\pi} (1 - \tau_-) f_{\tilde{P}}(\mathbf{x}) = (\tilde{\pi} - \tau_-) f_P(\mathbf{x}) + \tau_- (1 - \tilde{\pi}) f_{\tilde{Q}}(\mathbf{x}),$$

by using $f_{D_{\mathcal{X}}}(\mathbf{x}) = \tilde{\pi} f_{\tilde{P}}(\mathbf{x}) + (1 - \tilde{\pi}) f_{\tilde{Q}}(\mathbf{x})$. In a similar manner, we prove

$$(1 - \tilde{\pi})(1 - \tau_+) f_{\tilde{Q}}(\mathbf{x}) = (1 - \tilde{\pi} - \tau_+) f_Q(\mathbf{x}) + \tau_+ \tilde{\pi} f_{\tilde{P}}(\mathbf{x}),$$

and this lemma follows as desired. $\square$

Appendix B. Proof of Lemma 2

From Eqn. (1), we have $f_{D_{\mathcal{X}}}(\mathbf{x}) = \tilde{\pi} f_{\tilde{P}}(\mathbf{x}) + (1 - \tilde{\pi}) f_{\tilde{Q}}(\mathbf{x})$. This gives the first-order statistics w.r.t. the marginal distribution $D_{\mathcal{X}}$ as

$$\begin{aligned} \mu_{D_{\mathcal{X}}} &= E_{\mathbf{x} \sim D_{\mathcal{X}}}[\mathbf{x}] = \int_{\mathbf{x} \in \mathcal{X}} \mathbf{x} f_{D_{\mathcal{X}}}(\mathbf{x}) d\mathbf{x} = \int_{\mathbf{x} \in \mathcal{X}} \mathbf{x} \left(\tilde{\pi} f_{\tilde{P}}(\mathbf{x}) + (1 - \tilde{\pi}) f_{\tilde{Q}}(\mathbf{x}) \right) d\mathbf{x} \\ &= \tilde{\pi} \int_{\mathbf{x} \in \mathcal{X}} \mathbf{x} f_{\tilde{P}}(\mathbf{x}) d\mathbf{x} + (1 - \tilde{\pi}) \int_{\mathbf{x} \in \mathcal{X}} \mathbf{x} f_{\tilde{Q}}(\mathbf{x}) d\mathbf{x} = \tilde{\pi} \mu_{\tilde{P}} + (1 - \tilde{\pi}) \mu_{\tilde{Q}}, \end{aligned}$$

and the second-order statistics w.r.t. the marginal distribution $D_{\mathcal{X}}$ is given by

$$\begin{aligned}\Sigma_{D_{\mathcal{X}}} &= E_{\mathbf{x} \sim D_{\mathcal{X}}}[\mathbf{x}\mathbf{x}^T] = \int_{\mathbf{x} \in \mathcal{X}} \mathbf{x}\mathbf{x}^T f_{D_{\mathcal{X}}}(\mathbf{x}) d\mathbf{x} = \int_{\mathbf{x} \in \mathcal{X}} \mathbf{x}\mathbf{x}^T \left(\tilde{\pi} f_{\tilde{P}}(\mathbf{x}) + (1 - \tilde{\pi}) f_{\tilde{Q}}(\mathbf{x}) \right) d\mathbf{x} \\ &= \tilde{\pi} \int_{\mathbf{x} \in \mathcal{X}} \mathbf{x}\mathbf{x}^T f_{\tilde{P}}(\mathbf{x}) d\mathbf{x} + (1 - \tilde{\pi}) \int_{\mathbf{x} \in \mathcal{X}} \mathbf{x}\mathbf{x}^T f_{\tilde{Q}}(\mathbf{x}) d\mathbf{x} = \tilde{\pi} \Sigma_{\tilde{P}} + (1 - \tilde{\pi}) \Sigma_{\tilde{Q}}.\end{aligned}$$

Hence, we have the covariance matrix

$$\text{Cov}_{D_{\mathcal{X}}} = \Sigma_{D_{\mathcal{X}}} - \mu_{D_{\mathcal{X}}}[\mu_{D_{\mathcal{X}}}]^T = \tilde{\pi} \Sigma_{\tilde{P}} + (1 - \tilde{\pi}) \Sigma_{\tilde{Q}} - [\tilde{\pi} \mu_{\tilde{P}} + (1 - \tilde{\pi}) \mu_{\tilde{Q}}][\tilde{\pi} \mu_{\tilde{P}} + (1 - \tilde{\pi}) \mu_{\tilde{Q}}]^T,$$

which proves Eqn. (14).

From Eqn. (2), we have the first- and second-order statistics w.r.t. the corrupted class distribution $\tilde{P}$, respectively, as

$$\mu_{\tilde{P}} = \frac{\tilde{\pi} - \tau_-}{\tilde{\pi}(1 - \tau_-)} \mu_P + \frac{\tau_-(1 - \tilde{\pi})}{\tilde{\pi}(1 - \tau_-)} \mu_{\tilde{Q}} \quad \text{and} \quad \Sigma_{\tilde{P}} = \frac{\tilde{\pi} - \tau_-}{\tilde{\pi}(1 - \tau_-)} \Sigma_P + \frac{\tau_-(1 - \tilde{\pi})}{\tilde{\pi}(1 - \tau_-)} \Sigma_{\tilde{Q}}.$$

This gives the first- and second-order statistics of the ground-truth class distribution P without corruption, respectively, as

$$\mu_P = \frac{\tilde{\pi}(1 - \tau_-)}{\tilde{\pi} - \tau_-} \mu_{\tilde{P}} - \frac{\tau_-(1 - \tilde{\pi})}{\tilde{\pi} - \tau_-} \mu_{\tilde{Q}} \quad \text{and} \quad \Sigma_P = \frac{\tilde{\pi}(1 - \tau_-)}{\tilde{\pi} - \tau_-} \Sigma_{\tilde{P}} - \frac{\tau_-(1 - \tilde{\pi})}{\tilde{\pi} - \tau_-} \Sigma_{\tilde{Q}}.$$

In a similar manner, we obtain the first- and second-order statistics w.r.t. the ground-truth class distribution Q without corruption, from Eqn. (3), as follows:

$$\mu_Q = \frac{(1 - \tilde{\pi})(1 - \tau_+)}{1 - \tilde{\pi} - \tau_+} \mu_{\tilde{Q}} - \frac{\tilde{\pi} \tau_+}{1 - \tilde{\pi} - \tau_+} \mu_{\tilde{P}} \quad \text{and} \quad \Sigma_Q = \frac{(1 - \tilde{\pi})(1 - \tau_+)}{1 - \tilde{\pi} - \tau_+} \Sigma_{\tilde{Q}} - \frac{\tilde{\pi} \tau_+}{1 - \tilde{\pi} - \tau_+} \Sigma_{\tilde{P}}.$$

This completes the proof. $\square$

Appendix C. Proof of Theorem 1

We will present the detailed proof for the estimation of noise proportion τ_-^* , and estimate τ_+^* similarly. It is easy to observe that

$$\begin{aligned}\tau_-^* &= \max \{ \tau \in [0, \tilde{\pi}]: F(\tau) \geq 0 \} \\ &= \max \left\{ \tau \in [0, \tilde{\pi}]: \mathbf{w}^T F(\tau) \mathbf{w} \geq 0 \text{ for every } \mathbf{w} \in \mathbb{R}^d \text{ with } \|\mathbf{w}\| = 1 \right\}.\end{aligned}$$

Given any $\mathbf{w} \in \mathbb{R}^d$ with $\|\mathbf{w}\| = 1$, we will prove that there exists a unique $\tau(\mathbf{w}) \in [0, \tilde{\pi}]$ such that, for any $\tau \leq \tau(\mathbf{w})$,

$$g(\tau, \mathbf{w}) = (\tilde{\pi} - \tau)^2 \mathbf{w}^T \left(\Sigma_P(\tau) - \mu_P(\tau)[\mu_P(\tau)]^T \right) \mathbf{w} \geq 0.$$

We further simplify $g(\tau, \mathbf{w})$ as

$$g(\tau, \mathbf{w}) = a\tau^2 - b\tau + c,$$

where

$$\begin{aligned}a &= \mathbf{w}^T \left(\tilde{\pi} \Sigma_{\tilde{P}} + (1 - \tilde{\pi}) \Sigma_{\tilde{Q}} - [\tilde{\pi} \mu_{\tilde{P}} + (1 - \tilde{\pi}) \mu_{\tilde{Q}}][\tilde{\pi} \mu_{\tilde{P}} + (1 - \tilde{\pi}) \mu_{\tilde{Q}}]^T \right) \mathbf{w}, \\ b &= \tilde{\pi} \mathbf{w}^T \left((1 + \tilde{\pi}) \Sigma_{\tilde{P}} + (1 - \tilde{\pi}) \Sigma_{\tilde{Q}} - 2\tilde{\pi} \mu_{\tilde{P}} \mu_{\tilde{P}}^T - 2(1 - \tilde{\pi}) \mu_{\tilde{P}} \mu_{\tilde{Q}}^T \right) \mathbf{w}, \\ c &= \tilde{\pi}^2 \mathbf{w}^T \left(\Sigma_{\tilde{P}} - \mu_{\tilde{P}} \mu_{\tilde{P}}^T \right) \mathbf{w}.\end{aligned}$$

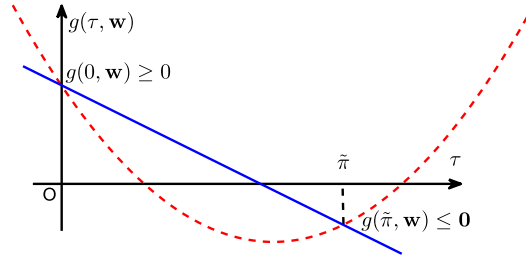
It is easy to observe $a \geq 0$ from the positive semi-definiteness of covariance matrices by Eqn. (14). We also have

$$\begin{aligned}g(0, \mathbf{w}) &= \tilde{\pi}^2 \mathbf{w}^T \left(\Sigma_{\tilde{P}} - \mu_{\tilde{P}} \mu_{\tilde{P}}^T \right) \mathbf{w} \geq 0, \\ g(\tilde{\pi}, \mathbf{w}) &= -\tilde{\pi}(1 - \tilde{\pi}) \mathbf{w}^T [\mu_{\tilde{P}} - \mu_{\tilde{Q}}][\mu_{\tilde{P}} - \mu_{\tilde{Q}}]^T \mathbf{w} \leq 0.\end{aligned}$$

As shown in Fig. 3, the function $g(\tau, \mathbf{w})$ can be illustrated as a (blue and solid) line if $a = 0$, and as a (red and dashed) parabolic curve if $a > 0$. Therefore, there exists a unique $\tau(\mathbf{w}) \in [0, \tilde{\pi}]$ such that $g(\tau) \geq 0$ for $\tau \leq \tau(\mathbf{w})$. We finally get

$$\tau_-^* = \min_{\mathbf{w} \in \mathbb{R}^d, \|\mathbf{w}\|=1} \{ \tau(\mathbf{w}) \} = \min_{\mathbf{w} \in \mathbb{R}^d, \|\mathbf{w}\|=1} \{ \max \{ \tau \in [0, \tilde{\pi}]: g(\tau, \mathbf{w}) \geq 0 \} \},$$

which completes the proof. $\square$

Fig. 3. The function $g(\tau, \mathbf{w})$.

Appendix D. Proof of Theorem 3

We present a detailed version of Theorem 3 as follows:

Theorem 4. Assume $\tilde{\pi} \in (0, 1)$, $\hat{\pi} \in (0, 1)$, $\text{Cov}_{\mathcal{D}_{\mathcal{X}}} > 0$ and $\widehat{\text{Cov}}_{\mathcal{D}_{\mathcal{X}}} > 0$. For any $\delta \in (0, 1)$ and for instance space $\mathcal{X} = \{\mathbf{x} \in \mathbb{R}^d : \|\mathbf{x}\| \leq B\}$, the following holds with probability at least $1 - \delta$ over the corrupted training sample $\tilde{S}_n$:

$$|\hat{\tau}_-^* - \tau_-^*| \leq \frac{\Pi_1}{\lambda_{\min}^2(\text{Cov}_{\mathcal{D}_{\mathcal{X}}})} + \frac{\Pi_1}{\lambda_{\min}^2(\widehat{\text{Cov}}_{\mathcal{D}_{\mathcal{X}}})} + \frac{\Pi_2}{\lambda_{\min}(\widehat{\text{Cov}}_{\mathcal{D}_{\mathcal{X}}})\lambda_{\min}(\text{Cov}_{\mathcal{D}_{\mathcal{X}}})}, \quad (24)$$

where

$$\begin{aligned} \Pi_1 &= \frac{6B^4}{n_{\min}^2} + \frac{10B^3}{n_{\min}} + B^3 \sqrt{\frac{10B^2}{n_{\min}^2} + \frac{12B}{n_{\min}}}, \\ \Pi_2 &= B^4 \sqrt{\Omega} \left(\frac{24}{\sqrt{n}} + \frac{25}{\sqrt{n_+}} + \frac{22}{\sqrt{n_-}} \right) + B^4 \sqrt[4]{\Omega} \left(\frac{5}{\sqrt[4]{n}} + \frac{6}{\sqrt[4]{n_+}} + \frac{4}{\sqrt[4]{n_-}} \right), \end{aligned}$$

with $n_{\min} = \min(n_+, n_-)$ and $\Omega = d \ln(3n_{\min}) + \ln(1/\delta)$. If there exists some $\rho \geq 0$ such that $\hat{\tau}_-^* \leq \hat{\pi} - \rho$ and $\tau_-^* \leq \tilde{\pi} - \rho$, then the following holds with probability at least $1 - \delta$ over the corrupted training sample $\tilde{S}_n$:

$$|\hat{\tau}_-^* - \tau_-^*| \leq \frac{\Pi'_1}{\lambda_{\min}^2(\text{Cov}_{\mathcal{D}_{\mathcal{X}}})} + \frac{\Pi'_2}{\lambda_{\min}^2(\widehat{\text{Cov}}_{\mathcal{D}_{\mathcal{X}}})} + \frac{\Pi'_3}{\lambda_{\min}(\widehat{\text{Cov}}_{\mathcal{D}_{\mathcal{X}}})\lambda_{\min}(\text{Cov}_{\mathcal{D}_{\mathcal{X}}})}, \quad (25)$$

where

$$\begin{aligned} \Pi'_1 &= \frac{6B^4}{n_{\min}^2} + \frac{10B^3}{n_{\min}} + \frac{B^4}{\rho \lambda_{\min}(\text{Cov}_{\mathcal{D}_{\mathcal{X}}})} \left(\frac{10B^2}{n_{\min}^2} + \frac{12B}{n_{\min}} \right), \\ \Pi'_2 &= \frac{6B^4}{n_{\min}^2} + \frac{10B^3}{n_{\min}} + \frac{B^4}{\rho \lambda_{\min}(\widehat{\text{Cov}}_{\mathcal{D}_{\mathcal{X}}})} \left(\frac{10B^2}{n_{\min}^2} + \frac{12B}{n_{\min}} \right), \\ \Pi'_3 &= B^4 \sqrt{\Omega} \left(\frac{24}{\sqrt{n}} + \frac{25}{\sqrt{n_+}} + \frac{22}{\sqrt{n_-}} \right) + \frac{B^4 \sqrt{\Omega}/\rho}{\lambda_{\min}(\text{Cov}_{\mathcal{D}_{\mathcal{X}}})} \left(\frac{46}{\sqrt{n}} + \frac{58}{\sqrt{n_+}} + \frac{26}{\sqrt{n_-}} \right). \end{aligned}$$

We first present some helpful lemmas before the proof of Theorem 4.

Lemma 3 (Hoeffding bounds [26]). Let $X_1, X_2, \dots, X_n$ be n independent random variables with $X_i \in [a_i, b_i]$. For any $\epsilon > 0$, we have

$$\Pr \left[\left| \frac{1}{n} \sum_{i=1}^n X_i - \frac{1}{n} \sum_{i=1}^n E[X_i] \right| \geq \epsilon \right] \leq \exp \left(\frac{-2n^2 \epsilon^2}{\sum_{i=1}^n (b_i - a_i)^2} \right).$$

Lemma 4. Let $\mathcal{X} = \{\mathbf{x} \in \mathbb{R}^d : \|\mathbf{x}\| \leq B\}$ and fix $\mathbf{w} \in \mathbb{R}^d$ with $\|\mathbf{w}\| = 1$. For every $\delta > 0$, the followings hold with probability at least $1 - 5\delta$ over the corrupted training sample $\tilde{S}_n$:

$$\begin{aligned} |\tilde{\pi} - \hat{\pi}| &\leq \sqrt{\ln(2/\delta)/2n}, \\ |\mathbf{w}^\top (\hat{\Sigma}_{\tilde{\mathbf{p}}} - \Sigma_{\tilde{\mathbf{p}}}) \mathbf{w}| &\leq B^2 \sqrt{\ln(2/\delta)/2n_+}, \\ |\mathbf{w}^\top (\hat{\Sigma}_{\tilde{\mathbf{q}}} - \Sigma_{\tilde{\mathbf{q}}}) \mathbf{w}| &\leq B^2 \sqrt{\ln(2/\delta)/2n_-}, \end{aligned}$$

$$\begin{aligned}
|\mathbf{w}^\top (\hat{\mu}_{\bar{p}} \hat{\mu}_{\bar{p}}^\top - \mu_{\bar{p}} \mu_{\bar{p}}^\top) \mathbf{w}| &\leq 2B^2 \sqrt{2 \ln(2/\delta)/n_+}, \\
|\mathbf{w}^\top (\hat{\mu}_{\bar{Q}} \hat{\mu}_{\bar{Q}}^\top - \mu_{\bar{Q}} \mu_{\bar{Q}}^\top) \mathbf{w}| &\leq 2B^2 \sqrt{2 \ln(2/\delta)/n_-}, \\
|\mathbf{w}^\top (\hat{\mu}_{\bar{p}} \hat{\mu}_{\bar{Q}}^\top - \mu_{\bar{p}} \mu_{\bar{Q}}^\top) \mathbf{w}| &\leq B^2 \sqrt{2 \ln(2/\delta)/n_+} + B^2 \sqrt{2 \ln(2/\delta)/n_-}.
\end{aligned}$$

Proof. It is easy to get $E[\hat{\pi}] = \tilde{\pi}$ from $\hat{\pi} = \sum_{i=1}^n I[\tilde{y}_i = +1]/n$ and $I[\tilde{y}_i = +1] \in \{0, 1\}$, where $I[\cdot]$ denotes the indicator function, which returns 1 if the argument is true, and 0 otherwise. By using the Hoeffding bounds from Lemma 3, we have

$$\Pr_{\tilde{S}_n} [|\hat{\pi} - \tilde{\pi}| \geq \epsilon] \leq 2 \exp(-2n\epsilon^2).$$

This follows that $|\hat{\pi} - \tilde{\pi}| \leq \sqrt{\ln(2/\delta)}/\sqrt{2n}$ by setting $\delta = 2 \exp(-2n\epsilon^2)$. We also have

$$\begin{aligned}
\left| \mathbf{w}^\top (\hat{\mu}_{\bar{p}} \hat{\mu}_{\bar{p}}^\top - \mu_{\bar{p}} \mu_{\bar{p}}^\top) \mathbf{w} \right| &= \left| \langle \mathbf{w}, \hat{\mu}_{\bar{p}} \rangle^2 - \langle \mathbf{w}, \mu_{\bar{p}} \rangle^2 \right| \\
&\leq \left| \langle \mathbf{w}, \hat{\mu}_{\bar{p}} \rangle - \langle \mathbf{w}, \mu_{\bar{p}} \rangle \right| \left(\left| \langle \mathbf{w}, \hat{\mu}_{\bar{p}} \rangle \right| + \left| \langle \mathbf{w}, \mu_{\bar{p}} \rangle \right| \right) \\
&\leq 2B \left| \langle \mathbf{w}, \hat{\mu}_{\bar{p}} \rangle - \langle \mathbf{w}, \mu_{\bar{p}} \rangle \right|,
\end{aligned}$$

where we use $\|\hat{\mu}_{\bar{p}}\| \leq B$, $\|\mu_{\bar{p}}\| \leq B$, $\|\mathbf{w}\| = 1$ and the Cauchy-Schwarz inequality to get

$$\left| \langle \mathbf{w}, \hat{\mu}_{\bar{p}} \rangle \right| \leq \|\mathbf{w}\| \|\hat{\mu}_{\bar{p}}\| \leq B \quad \text{and} \quad \left| \langle \mathbf{w}, \mu_{\bar{p}} \rangle \right| \leq \|\mathbf{w}\| \|\mu_{\bar{p}}\| \leq B.$$

We further have $E[\langle \mathbf{w}, \hat{\mu}_{\bar{p}} \rangle] = \langle \mathbf{w}, \mu_{\bar{p}} \rangle$ and $|\langle \mathbf{x}_i, \mathbf{w} \rangle| \leq B$. By using Lemma 3 again, we have

$$\Pr \left[\left| \langle \mathbf{w}, \hat{\mu}_{\bar{p}} \rangle - \langle \mathbf{w}, \mu_{\bar{p}} \rangle \right| \geq \epsilon \right] \leq 2 \exp(n_+ \epsilon^2 / 2B^2).$$

By setting $\delta = 2 \exp(n_+ \epsilon^2 / 2B^2)$, we have

$$\left| \langle \mathbf{w}, \hat{\mu}_{\bar{p}} \rangle - \langle \mathbf{w}, \mu_{\bar{p}} \rangle \right| \leq B \sqrt{2 \ln(2/\delta)/n_+},$$

and this follows that

$$\left| \mathbf{w}^\top (\hat{\mu}_{\bar{p}} \hat{\mu}_{\bar{p}}^\top - \mu_{\bar{p}} \mu_{\bar{p}}^\top) \mathbf{w} \right| \leq 2B^2 \sqrt{2 \ln(2/\delta)/n_+}.$$

Similarly, we can bound $|\mathbf{w}^\top (\hat{\mu}_{\bar{Q}} \hat{\mu}_{\bar{Q}}^\top - \mu_{\bar{Q}} \mu_{\bar{Q}}^\top) \mathbf{w}| \leq 2B^2 \sqrt{2 \ln(2/\delta)/\sqrt{n_-}}$. In addition, we have

$$\begin{aligned}
\left| \mathbf{w}^\top (\hat{\mu}_{\bar{p}} \hat{\mu}_{\bar{Q}}^\top - \mu_{\bar{p}} \mu_{\bar{Q}}^\top) \mathbf{w} \right| &= \left| \mathbf{w}^\top (\hat{\mu}_{\bar{p}} \hat{\mu}_{\bar{Q}}^\top - \hat{\mu}_{\bar{p}} \mu_{\bar{Q}}^\top + \hat{\mu}_{\bar{p}} \mu_{\bar{Q}}^\top - \mu_{\bar{p}} \mu_{\bar{Q}}^\top) \mathbf{w} \right| \\
&\leq \left| \mathbf{w}^\top \hat{\mu}_{\bar{p}} (\hat{\mu}_{\bar{Q}}^\top - \mu_{\bar{Q}}^\top) \mathbf{w} + \mathbf{w}^\top (\hat{\mu}_{\bar{p}} - \mu_{\bar{p}}) \mu_{\bar{Q}}^\top \mathbf{w} \right| \\
&\leq B \left| \langle \hat{\mu}_{\bar{Q}} - \mu_{\bar{Q}}, \mathbf{w} \rangle \right| + B \left| \langle \mathbf{w}, \hat{\mu}_{\bar{p}} - \mu_{\bar{p}} \rangle \right| \\
&\leq B^2 \sqrt{2 \ln(2/\delta)} (1/\sqrt{n_+} + 1/\sqrt{n_-}).
\end{aligned}$$

Finally, we have $E[\hat{\Sigma}_{\bar{p}}] = \Sigma_{\bar{p}}$ and $0 \leq \mathbf{w}^\top \mathbf{x}_i \mathbf{x}_i^\top \mathbf{w} \leq B^2$. By using Lemma 3 again, we have

$$\Pr \left[\left| \mathbf{w}^\top (\hat{\Sigma}_{\bar{p}} - \Sigma_{\bar{p}}) \mathbf{w} \right| \geq \epsilon \right] \leq 2 \exp(-2n_+ \epsilon^2 / B^4).$$

By setting $\delta = 2 \exp(-2n_+ \epsilon^2 / B^4)$, the following holds with probability at least $1 - \delta$

$$|\mathbf{w}^\top (\hat{\Sigma}_{\bar{p}} - \Sigma_{\bar{p}}) \mathbf{w}| \leq B^2 \sqrt{\ln(2/\delta)/\sqrt{2n_+}}.$$

Similarly, we can bound $|\mathbf{w}^\top (\hat{\Sigma}_{\bar{Q}} - \Sigma_{\bar{Q}}) \mathbf{w}| \leq B^2 \sqrt{\ln(2/\delta)/\sqrt{2n_-}}$, and the lemma follows. $\square$

Lemma 5. Let $\mathcal{X} = \{\mathbf{x} \in \mathbb{R}^d : \|\mathbf{x}\| \leq B\}$ and fix $\mathbf{w} \in \mathbb{R}^d$ with $\|\mathbf{w}\| = 1$. For every $\delta > 0$, the followings hold with probability at least $1 - 5\delta$ over the corrupted training sample $\tilde{S}_n$:

$$\begin{aligned}
|a - \hat{a}| &\leq 5B^2 \sqrt{\ln(2/\delta)/n} + 5B^2 \sqrt{\ln(2/\delta)/n_+} + 5B^2 \sqrt{\ln(2/\delta)/n_-}, \\
|b - \hat{b}| &\leq 8B^2 \sqrt{\ln(2/\delta)/n} + 10B^2 \sqrt{\ln(2/\delta)/n_+} + 4B^2 \sqrt{\ln(2/\delta)/n_-}, \\
|c - \hat{c}| &\leq 2B^2 \sqrt{\ln(2/\delta)/n} + 4B^2 \sqrt{\ln(2/\delta)/n_+},
\end{aligned}$$

where

$$\begin{aligned}
a &= \mathbf{w}^\top \left(\tilde{\pi} \Sigma_{\tilde{p}} + (1 - \tilde{\pi}) \Sigma_{\tilde{Q}} - [\tilde{\pi} \mu_{\tilde{p}} + (1 - \tilde{\pi}) \mu_{\tilde{Q}}] [\tilde{\pi} \mu_{\tilde{p}} + (1 - \tilde{\pi}) \mu_{\tilde{Q}}]^\top \right) \mathbf{w}, \\
\hat{a} &= \mathbf{w}^\top \left(\hat{\pi} \hat{\Sigma}_{\tilde{p}} + (1 - \hat{\pi}) \hat{\Sigma}_{\tilde{Q}} - [\hat{\pi} \hat{\mu}_{\tilde{p}} + (1 - \hat{\pi}) \mu_{\tilde{Q}}] [\hat{\pi} \hat{\mu}_{\tilde{p}} + (1 - \hat{\pi}) \mu_{\tilde{Q}}]^\top \right) \mathbf{w}, \\
b &= \tilde{\pi} \mathbf{w}^\top \left((1 + \tilde{\pi}) \Sigma_{\tilde{p}} + (1 - \tilde{\pi}) \Sigma_{\tilde{Q}} - 2\tilde{\pi} \mu_{\tilde{p}} \mu_{\tilde{p}}^\top - 2(1 - \tilde{\pi}) \mu_{\tilde{p}} \mu_{\tilde{Q}}^\top \right) \mathbf{w}, \\
\hat{b} &= \hat{\pi} \mathbf{w}^\top \left((1 + \hat{\pi}) \hat{\Sigma}_{\tilde{p}} + (1 - \hat{\pi}) \hat{\Sigma}_{\tilde{Q}} - 2\hat{\pi} \hat{\mu}_{\tilde{p}} \hat{\mu}_{\tilde{p}}^\top - 2(1 - \hat{\pi}) \hat{\mu}_{\tilde{p}} \mu_{\tilde{Q}}^\top \right) \mathbf{w}, \\
c &= \tilde{\pi}^2 \mathbf{w}^\top \left(\Sigma_{\tilde{p}} - \mu_{\tilde{p}} \mu_{\tilde{p}}^\top \right) \mathbf{w}, \\
\hat{c} &= \hat{\pi}^2 \mathbf{w}^\top \left(\hat{\Sigma}_{\tilde{p}} - \hat{\mu}_{\tilde{p}} \hat{\mu}_{\tilde{p}}^\top \right) \mathbf{w}.
\end{aligned}$$

Proof. We first have $|a| \leq B^2$, $|\hat{a}| \leq B^2$, $|c| \leq B^2$, $|\hat{c}| \leq B^2$, $|b| \leq 4B^2$ and $|\hat{b}| \leq 4B^2$, and this holds that

$$\begin{aligned}
|c - \hat{c}| &\leq |\tilde{\pi}^2 - \hat{\pi}^2| |\mathbf{w}^\top (\Sigma_{\tilde{p}} - \mu_{\tilde{p}} \mu_{\tilde{p}}^\top) \mathbf{w}| + \hat{\pi}^2 |\mathbf{w}^\top (\Sigma_{\tilde{p}} - \mu_{\tilde{p}} \mu_{\tilde{p}}^\top - \hat{\Sigma}_{\tilde{p}} + \hat{\mu}_{\tilde{p}} \hat{\mu}_{\tilde{p}}^\top) \mathbf{w}| \\
&\leq 2B^2 |\tilde{\pi} - \hat{\pi}| + |\mathbf{w}^\top (\Sigma_{\tilde{p}} - \hat{\Sigma}_{\tilde{p}}) \mathbf{w}| + |\mathbf{w}^\top (\mu_{\tilde{p}} \mu_{\tilde{p}}^\top - \hat{\mu}_{\tilde{p}} \hat{\mu}_{\tilde{p}}^\top) \mathbf{w}| \\
&\leq 2B^2 \sqrt{\ln(2/\delta)/2n} + B^2 \sqrt{\ln(2/\delta)/2n_+} + 2B^2 \sqrt{2\ln(2/\delta)/n_+} \\
&\leq 2B^2 \sqrt{\ln(2/\delta)/n} + 4B^2 \sqrt{\ln(2/\delta)/n_+},
\end{aligned}$$

where the last inequality holds from Lemma 4 and simple calculations. We also have

$$\begin{aligned}
|b - \hat{b}| &\leq |\tilde{\pi}(1 + \tilde{\pi}) \mathbf{w}^\top \Sigma_{\tilde{p}} \mathbf{w} - \hat{\pi}(1 + \hat{\pi}) \mathbf{w}^\top \hat{\Sigma}_{\tilde{p}} \mathbf{w}| + |\tilde{\pi}(1 - \tilde{\pi}) \mathbf{w}^\top \Sigma_{\tilde{Q}} \mathbf{w} - \hat{\pi}(1 - \hat{\pi}) \mathbf{w}^\top \hat{\Sigma}_{\tilde{Q}} \mathbf{w}| \\
&\quad + 2|\tilde{\pi}^2 \mathbf{w}^\top \mu_{\tilde{p}} \mu_{\tilde{p}}^\top \mathbf{w} - \hat{\pi}^2 \mathbf{w}^\top \hat{\mu}_{\tilde{p}} \hat{\mu}_{\tilde{p}}^\top \mathbf{w}| + 2|\tilde{\pi}(1 - \tilde{\pi}) \mathbf{w}^\top \mu_{\tilde{p}} \mu_{\tilde{Q}}^\top \mathbf{w} - \hat{\pi}(1 - \hat{\pi}) \mathbf{w}^\top \hat{\mu}_{\tilde{p}} \mu_{\tilde{Q}}^\top \mathbf{w}| \\
&\leq 8B^2 \sqrt{\ln(2/\delta)/n} + 10B^2 \sqrt{\ln(2/\delta)/n_+} + 4B^2 \sqrt{\ln(2/\delta)/n_-}.
\end{aligned}$$

In a similar manner, we have

$$|a - \hat{a}| \leq 5B^2 \sqrt{\ln(2/\delta)/n} + 5B^2 \sqrt{\ln(2/\delta)/n_+} + 5B^2 \sqrt{\ln(2/\delta)/n_-},$$

which completes the proof. $\square$

Lemma 6. For constants $A_0 > 0$, $\hat{A}_0 > 0$, A_1 , B_1 , and C_1 , let $a \in [A_0, A_1]$, $\hat{a} \in [\hat{A}_0, A_1]$, $b, \hat{b} \in [0, B_1]$, and $c, \hat{c} \in [0, C_1]$. We have

$$\left| \frac{b - \sqrt{b^2 - 4ac}}{2a} - \frac{\hat{b} - \sqrt{\hat{b}^2 - 4\hat{a}\hat{c}}}{2\hat{a}} \right| \leq \frac{2B_1\gamma_a + A_1\gamma_b + A_1\sqrt{2B_1\gamma_b + 4A_1\gamma_c + 4C_1\gamma_a}}{2A_0\hat{A}_0},$$

if $|a - \hat{a}| \leq \gamma_a$, $|b - \hat{b}| \leq \gamma_b$, $|c - \hat{c}| \leq \gamma_c$, $b^2 \geq 4ac$ and $\hat{b}^2 \geq 4\hat{a}\hat{c}$. Moreover, if $\Delta = b^2 - 4ac > 0$, then we have

$$\left| \frac{b - \sqrt{b^2 - 4ac}}{2a} - \frac{\hat{b} - \sqrt{\hat{b}^2 - 4\hat{a}\hat{c}}}{2\hat{a}} \right| \leq \frac{2B_1\gamma_a + A_1\gamma_b + A_1(2B_1\gamma_b + 4A_1\gamma_c + 4C_1\gamma_a)/\sqrt{\Delta}}{2A_0\hat{A}_0}.$$

Proof. We first have

$$\begin{aligned}
&\left| \frac{b - \sqrt{b^2 - 4ac}}{2a} - \frac{\hat{b} - \sqrt{\hat{b}^2 - 4\hat{a}\hat{c}}}{2\hat{a}} \right| \\
&\leq \frac{|b||a - \hat{a}| + |a - \hat{a}|\sqrt{b^2 - 4ac} + |a||b - \hat{b}| + |a|\sqrt{b^2 - 4ac} - \sqrt{\hat{b}^2 - 4\hat{a}\hat{c}}|}{2a\hat{a}} \\
&\leq \frac{2|b||a - \hat{a}| + |a||b - \hat{b}| + |a|\sqrt{b^2 - 4ac} - \sqrt{\hat{b}^2 - 4\hat{a}\hat{c}}|}{2A_0\hat{A}_0} \\
&\leq \frac{2B_1\gamma_a + A_1\gamma_b + A_1|\sqrt{b^2 - 4ac} - \sqrt{\hat{b}^2 - 4\hat{a}\hat{c}}|}{2A_0\hat{A}_0}.
\end{aligned}$$

By using the formula $|\sqrt{x} - \sqrt{x'}| \leq \sqrt{|x - x'|}$ for $x \geq 0, x' \geq 0$, it holds that

$$\left| \sqrt{b^2 - 4ac} - \sqrt{\hat{b}^2 - 4\hat{a}\hat{c}} \right| \leq \sqrt{|b^2 - \hat{b}^2| + 4|ac - \hat{a}\hat{c}|} \leq \sqrt{2B_1\gamma_b + 4A_1\gamma_c + 4C_1\gamma_a}.$$

If $\Delta = b^2 - 4ac > 0$, then we have

$$\left| \sqrt{b^2 - 4ac} - \sqrt{\hat{b}^2 - 4\hat{a}\hat{c}} \right| \leq \frac{|b^2 - \hat{b}^2| + 4|ac - \hat{a}\hat{c}|}{\sqrt{b^2 - 4ac} + \sqrt{\hat{b}^2 - 4\hat{a}\hat{c}}} \leq \frac{2B_1\gamma_b + 4A_1\gamma_c + 4C_1\gamma_a}{\sqrt{\Delta}}.$$

This completes the proof. $\square$

Lemma 7. For $\epsilon \in (0, 1)$, if $\|\mathbf{w} - \mathbf{w}'\| \leq \epsilon$, then we have

$$\left| \frac{b - \sqrt{b^2 - 4ac}}{2a} - \frac{b' - \sqrt{b'^2 - 4a'c'}}{2a'} \right| \leq \frac{6B^4\epsilon^2 + 10B^3\epsilon + B^3\sqrt{10B^2\epsilon^2 + 12B\epsilon}}{\lambda_{\min}^2(\text{Cov}_{\mathcal{D}_{\mathcal{X}}})},$$

where a, b, c are defined in Lemma 5, and

$$\begin{aligned} a' &= [\mathbf{w}']^\top (\tilde{\pi} \Sigma_{\tilde{\mathbf{p}}} + (1 - \tilde{\pi}) \Sigma_{\tilde{\mathbf{Q}}} - [\tilde{\pi} \mu_{\tilde{\mathbf{p}}} + (1 - \tilde{\pi}) \mu_{\tilde{\mathbf{Q}}}] [\tilde{\pi} \mu_{\tilde{\mathbf{p}}} + (1 - \tilde{\pi}) \mu_{\tilde{\mathbf{Q}}}]^\top) \mathbf{w}', \\ b' &= \tilde{\pi} [\mathbf{w}']^\top \left((1 + \tilde{\pi}) \Sigma_{\tilde{\mathbf{p}}} + (1 - \tilde{\pi}) \Sigma_{\tilde{\mathbf{Q}}} - 2\tilde{\pi} \mu_{\tilde{\mathbf{p}}} \mu_{\tilde{\mathbf{p}}}^\top - 2(1 - \tilde{\pi}) \mu_{\tilde{\mathbf{p}}} \mu_{\tilde{\mathbf{Q}}}^\top \right) \mathbf{w}', \\ c' &= \tilde{\pi}^2 [\mathbf{w}']^\top \left(\Sigma_{\tilde{\mathbf{p}}} - \mu_{\tilde{\mathbf{p}}} \mu_{\tilde{\mathbf{p}}}^\top \right) \mathbf{w}'. \end{aligned}$$

Moreover, if $b^2 - 4ac \geq \Delta$, then we have

$$\left| \frac{b - \sqrt{b^2 - 4ac}}{2a} - \frac{b' - \sqrt{b'^2 - 4a'c'}}{2a'} \right| \leq \frac{6B^4\epsilon^2 + 10B^3\epsilon + B^4(10B^2\epsilon^2 + 12B\epsilon)/\sqrt{\Delta}}{\lambda_{\min}^2(\text{Cov}_{\mathcal{D}_{\mathcal{X}}})}.$$

Proof. We first have $a, a' \in [\lambda_{\min}(\text{Cov}_{\mathcal{D}_{\mathcal{X}}}), B^2]$, $b, b' \in [0, 4B^2]$ and $c, c' \in [0, B^2]$. This follows

$$\begin{aligned} |a - a'| &= |\mathbf{w}^\top \text{Cov}_{\mathcal{D}_{\mathcal{X}}} \mathbf{w} - [\mathbf{w}']^\top \text{Cov}_{\mathcal{D}_{\mathcal{X}}} \mathbf{w}'| \\ &\leq |(\mathbf{w} - \mathbf{w}')^\top \text{Cov}_{\mathcal{D}_{\mathcal{X}}} (\mathbf{w} - \mathbf{w}')| + 2|\mathbf{w}^\top \text{Cov}_{\mathcal{D}_{\mathcal{X}}} (\mathbf{w} - \mathbf{w}')| \leq B^2\epsilon^2 + 2B\epsilon. \end{aligned}$$

Similarly, we have $|b - b'| \leq 4B^2\epsilon^2 + 4B\epsilon$ and $|c - c'| \leq B^2\epsilon^2 + 2B\epsilon$. This completes the proof by using Lemma 6 and simple calculations. $\square$

Proof of Theorem 4. Similarly to the proof of Theorem 1 in Appendix C, we first write

$$g(\tau, \mathbf{w}) = (\tilde{\pi} - \tau)^2 \mathbf{w}^\top F(\tau) \mathbf{w} = a\tau^2 - b\tau + c,$$

for $\mathbf{w} \in \mathbb{R}^d$ with $\|\mathbf{w}\| = 1$, where a, b and c are defined in Lemma 5. It is easy to get $g(0, \mathbf{w}) \geq 0$ and $g(\tilde{\pi}, \mathbf{w}) \leq 0$, and we have $b \in [0, 4B^2]$, $c \in [0, B^2]$, $a = \mathbf{w}^\top \text{Cov}_{\mathcal{D}_{\mathcal{X}}} \mathbf{w} \in [\lambda_{\min}(\text{Cov}_{\mathcal{D}_{\mathcal{X}}}), B^2]$ and $b^2 - 4ac \geq 0$, where $\lambda_{\min}(\text{Cov}_{\mathcal{D}_{\mathcal{X}}}) > 0$ from $\text{Cov}_{\mathcal{D}_{\mathcal{X}}} \succ 0$. Then, there exists a unique

$$\tau(\mathbf{w}) = (b - \sqrt{b^2 - 4ac})/2a,$$

such that $g(\tau(\mathbf{w}), \mathbf{w}) = 0$ and $\tau(\mathbf{w}) \in [0, \tilde{\pi}]$. From the proof of Theorem 1, we have

$$\tau_-^* = \min_{\|\mathbf{w}\|=1} \{\tau(\mathbf{w})\} = \min_{\|\mathbf{w}\|=1} \left\{ \left(b - \sqrt{b^2 - 4ac} \right) / 2a \right\}.$$

Similarly, we write

$$\hat{g}(\tau, \mathbf{w}) = (\hat{\pi} - \tau)^2 \mathbf{w}^\top \hat{F}(\tau) \mathbf{w} = \hat{a}\tau^2 - \hat{b}\tau + \hat{c},$$

where $\hat{a}, \hat{b}$ and $\hat{c}$ are defined in Lemma 5. Similarly, we have $\hat{g}(0, \mathbf{w}) \geq 0$, $\hat{g}(\hat{\pi}, \mathbf{w}) \leq 0$, $\hat{b} \in [0, 4B^2]$, $\hat{c} \in [0, B^2]$, $\hat{a} = \mathbf{w}^\top \widehat{\text{Cov}}_{\mathcal{D}_{\mathcal{X}}} \mathbf{w} \in [\lambda_{\min}(\widehat{\text{Cov}}_{\mathcal{D}_{\mathcal{X}}}), B^2]$ and $\hat{b}^2 - 4\hat{a}\hat{c} \geq 0$, where $\lambda_{\min}(\widehat{\text{Cov}}_{\mathcal{D}_{\mathcal{X}}}) > 0$ from the assumption $\widehat{\text{Cov}}_{\mathcal{D}_{\mathcal{X}}} \succ 0$. There exists a unique

$$\hat{\tau}(\mathbf{w}) = \left(\hat{b} - \sqrt{\hat{b}^2 - 4\hat{a}\hat{c}} \right) / 2\hat{a},$$

such that $\hat{g}(\hat{\tau}(\mathbf{w}), \mathbf{w}) = 0$ and $\hat{\tau}(\mathbf{w}) \in [0, \hat{\pi}]$. Similarly to the proof of Theorem 1, we have

$$\hat{\tau}_-^* = \min_{\|\mathbf{w}\|=1} \hat{\tau}(\mathbf{w}) = \min_{\|\mathbf{w}\|=1} \left\{ \left(\hat{b} - \sqrt{\hat{b}^2 - 4\hat{a}\hat{c}} \right) / 2\hat{a} \right\}.$$

Let $\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_m$ be a ϵ -cover of the set $\{\mathbf{w} \in \mathbb{R}^d: \|\mathbf{w}\| = 1\}$, and we have $m \leq (3/\epsilon)^d$ from [40, Lemma 4.16]. For any $\mathbf{w} \in \mathbb{R}^d$ with $\|\mathbf{w}\| = 1$, there exists some $\mathbf{w}_k$ ($k \in [m]$) such that $|\mathbf{w}_k - \mathbf{w}| \leq \epsilon$. This follows that

$$|\tau_-^* - \hat{\tau}_-^*| \leq |\tau(\mathbf{w}) - \hat{\tau}(\mathbf{w})| \leq |\tau(\mathbf{w}) - \tau(\mathbf{w}_k)| + |\hat{\tau}(\mathbf{w}_k) - \hat{\tau}(\mathbf{w})| + |\tau(\mathbf{w}_k) - \hat{\tau}(\mathbf{w}_k)|. \quad (26)$$

Based on Lemma 7, we have

$$|\tau(\mathbf{w}) - \tau(\mathbf{w}_k)| \leq \left(6B^4\epsilon^2 + 10B^3\epsilon + B^3\sqrt{10B^2\epsilon^2 + 12B\epsilon}\right) / \lambda_{\min}^2(\text{Cov}_{\mathcal{D}_{\mathcal{X}}}). \quad (27)$$

Similarly to the proof of Lemma 7, we have

$$|\hat{\tau}(\mathbf{w}_k) - \hat{\tau}(\mathbf{w})| \leq \left(6B^4\epsilon^2 + 10B^3\epsilon + B^3\sqrt{10B^2\epsilon^2 + 12B\epsilon}\right) / \lambda_{\min}^2(\widehat{\text{Cov}}_{\mathcal{D}_{\mathcal{X}}}). \quad (28)$$

By using the union bounds and combining with Lemmas 5 and 6, the following holds, for any $k \in [m]$, with probability at least $1 - \delta$ over the corrupted training sample $\tilde{S}_n$:

$$|\tau(\mathbf{w}_k) - \hat{\tau}(\mathbf{w}_k)| \leq \frac{B^4\sqrt{\Omega}\left(\frac{24}{\sqrt{n}} + \frac{25}{\sqrt{n_+}} + \frac{22}{\sqrt{n_-}}\right) + B^4\sqrt{\sqrt{\Omega}\left(\frac{23}{\sqrt{n}} + \frac{29}{\sqrt{n_+}} + \frac{13}{\sqrt{n_-}}\right)}}{\lambda_{\min}(\widehat{\text{Cov}}_{\mathcal{D}_{\mathcal{X}}})\lambda_{\min}(\text{Cov}_{\mathcal{D}_{\mathcal{X}}})}, \quad (29)$$

where $\Omega = d \ln(3/\epsilon) + \ln(1/\delta)$. Eqn. (24) holds by setting $\epsilon = 1/\min(n_+, n_-)$, and combining with Eqns. (26)-(29) and simple calculations.

On the other hand, if there exists some $\rho \geq 0$ s.t. $\hat{\tau}_-^* \leq \hat{\pi} - \rho$ and $\tau_-^* \leq \tilde{\pi} - \rho$, then we denote by

$$\mathcal{A}_1 = \left\{ \mathbf{w} \in \mathbb{R}^d: \|\mathbf{w}\| = 1, b^2 - 4ac \geq \rho^2\lambda_{\min}^2(\text{Cov}_{\mathcal{D}_{\mathcal{X}}}) \text{ and } \hat{b}^2 - 4\hat{a}\hat{c} \geq \rho^2\lambda_{\min}^2(\widehat{\text{Cov}}_{\mathcal{D}_{\mathcal{X}}}) \right\}.$$

For any $\mathbf{w} \notin \mathcal{A}_1$, we have $\tau(\mathbf{w}) \geq \tilde{\pi} - \rho$ from the facts that $g(\tau(\mathbf{w}), \mathbf{w}) = 0$, $g(\tilde{\pi}, \mathbf{w}) \leq 0$ and

$$\tilde{\pi} - \tau(\mathbf{w}) \leq \sqrt{b^2 - 4ac}/a \leq \sqrt{b^2 - 4ac}/\lambda_{\min}(\text{Cov}_{\mathcal{D}_{\mathcal{X}}}) \leq \rho.$$

This follows that $\tau_-^* = \min_{\mathbf{w} \in \mathcal{A}_1} \{\tau(\mathbf{w})\}$, and we have $\hat{\tau}_-^* = \min_{\mathbf{w} \in \mathcal{A}_1} \{\hat{\tau}(\mathbf{w})\}$ in a similar manner.

Let $\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_{m'}$ be a ϵ -cover of $\mathcal{A}_1$, and we have $m' \leq (3/\epsilon)^d$ from [40, Lemma 4.16]. For any $\mathbf{w} \in \mathcal{A}_1$, there exists some $\mathbf{w}_k$ ($k \in [m']$) such that $|\mathbf{w}_k - \mathbf{w}| \leq \epsilon$. Based on Lemma 7, we have

$$|\tau(\mathbf{w}) - \tau(\mathbf{w}_k)| \leq \frac{6B^4\epsilon^2 + 10B^3\epsilon + B^4(10B^2\epsilon^2 + 12B\epsilon)/\rho\lambda_{\min}(\text{Cov}_{\mathcal{D}_{\mathcal{X}}})}{\lambda_{\min}^2(\text{Cov}_{\mathcal{D}_{\mathcal{X}}})}. \quad (30)$$

Similarly, we have

$$|\hat{\tau}(\mathbf{w}_k) - \hat{\tau}(\mathbf{w})| \leq \frac{6B^4\epsilon^2 + 10B^3\epsilon + B^4(10B^2\epsilon^2 + 12B\epsilon)/\rho\lambda_{\min}(\widehat{\text{Cov}}_{\mathcal{D}_{\mathcal{X}}})}{\lambda_{\min}^2(\widehat{\text{Cov}}_{\mathcal{D}_{\mathcal{X}}})}. \quad (31)$$

By using the union bounds and combining with Lemmas 5 and 6, the following holds, for any $k \in [m']$, with probability at least $1 - \delta$ over the corrupted data $\tilde{S}_n$,

$$|\tau(\mathbf{w}_k) - \hat{\tau}(\mathbf{w}_k)| \leq \frac{B^4\sqrt{\Omega}\left(\frac{24}{\sqrt{n}} + \frac{25}{\sqrt{n_+}} + \frac{22}{\sqrt{n_-}}\right)}{\lambda_{\min}(\widehat{\text{Cov}}_{\mathcal{D}_{\mathcal{X}}})\lambda_{\min}(\text{Cov}_{\mathcal{D}_{\mathcal{X}}})} + \frac{B^4\sqrt{\Omega}\left(\frac{46}{\sqrt{n}} + \frac{58}{\sqrt{n_+}} + \frac{26}{\sqrt{n_-}}\right)}{\rho\lambda_{\min}(\text{Cov}_{\mathcal{D}_{\mathcal{X}}})\lambda_{\min}(\widehat{\text{Cov}}_{\mathcal{D}_{\mathcal{X}}})\lambda_{\min}(\text{Cov}_{\mathcal{D}_{\mathcal{X}}})} \quad (32)$$

with $\Omega = d \ln(3/\epsilon) + \ln(1/\delta)$. Eqn. (25) holds by setting $\epsilon = 1/\min(n_+, n_-)$, and combining with Eqns. (26), (30)-(32) and simple calculations. $\square$

Appendix E. Proof of Proposition 2

Let $\mathcal{X}_1$ be a continuous area in instance space $\mathcal{X}$ with conditional probability $\eta(\mathbf{x}) = 0$ for every $\mathbf{x} \in \mathcal{X}_1$. We have the ratio $\pi = 0$ of training positive instances in the area $\mathcal{X}_1$. Each negative label has been flipped independently with the ground-truth noise proportion τ_- , and we have the corrupted ratio $\hat{\pi} = \tau_-$ and the corrupted class distributions $\hat{\mathbf{P}}(\mathbf{x}) = \hat{\mathbf{Q}}(\mathbf{x})$ for $\mathbf{x} \in \mathcal{X}_1$. This follows that $\hat{\mu}_{\hat{\mathbf{P}}(\mathbf{x})} = \hat{\mu}_{\hat{\mathbf{Q}}(\mathbf{x})}$ and $\hat{\Sigma}_{\hat{\mathbf{P}}(\mathbf{x})} = \hat{\Sigma}_{\hat{\mathbf{Q}}(\mathbf{x})}$, and

$$\hat{F}(\tau) = \hat{\Sigma}_{\hat{\mathbf{Q}}(\mathbf{x})} - [\hat{\mu}_{\hat{\mathbf{Q}}(\mathbf{x})}][\hat{\mu}_{\hat{\mathbf{Q}}(\mathbf{x})}]^\top \geq 0,$$

where $\hat{F}(\tau)$ is defined by Eqn. (11). Based on Eqn. (13), we have

$$\hat{\tau}_-^* = \max\{\tau \in [0, \hat{\pi}]: \hat{F}(\tau) \geq 0\} = \tau_-.$$

Similarly, we prove $\hat{\tau}_+^* = \tau_+$ for the case $\eta(\mathbf{x}) = 1$ with each $\mathbf{x} \in \mathcal{X}_1$. This proposition follows. $\square$

References

- [1] D. Angluin, P. Laird, Learning from noisy examples, *Mach. Learn.* 4 (2) (1988) 343–370.
- [2] J. Aslam, S. Decatur, On the sample complexity of noise-tolerant learning, *Inf. Process. Lett.* 57 (4) (1996) 189–195.
- [3] J. Bekker, J. Davis, Estimating the class prior in positive and unlabeled data through decision tree induction, in: *Proceedings of the 32 AAAI Conference on Artificial Intelligence*, New Orleans, LA, 2018, pp. 2712–2719.
- [4] S. Ben-David, D. Pál, S. Shalev-Shwartz, Agnostic online learning, in: *Proceedings of the 22nd Conference on Learning Theory*, Montreal, Canada, 2009.
- [5] S. Bhadra, M. Hein, Correction of noisy labels via mutual consistency check, *Neurocomputing* 160 (2015) 34–52.
- [6] G. Blanchard, M. Flaska, G. Handy, S. Pozzi, C. Scott, Classification with asymmetric label noise: consistency and maximal denoising, *Electron. J. Stat.* 10 (2) (2016) 2780–2824.
- [7] G. Blanchard, G. Lee, C. Scott, Semi-supervised novelty detection, *J. Mach. Learn. Res.* 11 (2010) 2973–3009.
- [8] A. Blum, T. Mitchell, Combining labeled and unlabeled data with co-training, in: *Proceedings of the 11th Annual Conference on Computational Learning Theory*, Madison, WI, 1998, pp. 92–100.
- [9] D. Brabham, Crowdsourcing as a model for problem solving: an introduction and cases, *Convergence* 14 (1) (2008) 75–90.
- [10] C. Brodley, M. Friedl, Identifying mislabeled training data, *J. Artif. Intell. Res.* 11 (1999) 131–167.
- [11] N. Bshouty, S. Goldman, H. Mathias, S. Suri, H. Tamaki, Noise-tolerant distribution-free learning of general geometric concepts, *J. ACM* 45 (5) (1998) 863–890.
- [12] T. Bylander, Learning linear threshold functions in the presence of classification noise, in: *Proceedings of the 7th Annual Conference on Computational Learning Theory*, New Brunswick, NJ, 1994, pp. 340–347.
- [13] N. Cesa-Bianchi, E. Dichterman, P. Fischer, E. Shamir, H. Simon, Sample-efficient strategies for learning in the presence of noise, *J. ACM* 46 (5) (1999) 684–719.
- [14] N. Cesa-Bianchi, S. Shalev-Shwartz, O. Shamir, Online learning of noisy data, *IEEE Trans. Inf. Theory* 57 (12) (2011) 7907–7931.
- [15] K. Crammer, O. Dekel, J. Keshet, S. Shalev-Shwartz, Y. Singer, Online passive-aggressive algorithms, *J. Mach. Learn. Res.* 7 (2006) 551–585.
- [16] K. Crammer, A. Kulesza, M. Dredze, Adaptive regularization of weight vectors, *Mach. Learn.* 91 (2) (2013) 155–187.
- [17] V. Denchev, N. Ding, H. Neven, S. Vishwanathan, Robust classification with adiabatic quantum optimization, in: *Proceedings of the 29th International Conference on Machine Learning*, Edinburgh, Scotland, 2012, pp. 863–870.
- [18] F. Denis, PAC learning from positive statistical queries, in: *Proceedings of the 9th International Conference on Algorithmic Learning Theory*, Otzenhausen, Germany, 1998, pp. 112–126.
- [19] M. Dredze, K. Crammer, F. Pereira, Confidence-weighted linear classification, in: *Proceedings of the 25th International Conference on Machine Learning*, Helsinki, Finland, 2008, pp. 264–271.
- [20] M. du Plessis, G. Niu, M. Sugiyama, Analysis of Learning from Positive and Unlabeled Data, *Advances in Neural Information Processing Systems*, vol. 27, MIT Press, Cambridge, MA, 2014, pp. 703–711.
- [21] C. Elkan, K. Noto, Learning classifiers from only positive and unlabeled data, in: *Proceedings of the 14th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, Las Vegas, NV, 2008, pp. 213–220.
- [22] B. Fréna, M. Verleysen, Classification in the presence of label noise: a survey, *IEEE Trans. Neural Netw. Learn. Syst.* 25 (5) (2014) 845–869.
- [23] Y. Freund, A more robust boosting algorithm, CoRR/abstract, arXiv:0905.2138, 2009.
- [24] W. Gao, L. Wang, Y.-F. Li, Z.-H. Zhou, Risk minimization in the presence of label noise, in: *Proceedings of the 30 AAAI Conference on Artificial Intelligence*, Phoenix, AZ, 2016, pp. 1575–1581.
- [25] A. Ghosh, N. Manwani, P. Sastry, Making risk minimization tolerant to label noise, *Neurocomputing* 160 (2015) 93–107.
- [26] W. Hoeffding, Probability inequalities for sums of bounded random variables, *J. Am. Stat. Assoc.* 58 (301) (1963) 13–30.
- [27] A. Kalai, R. Servedio, Boosting in the presence of noise, *J. Comput. Syst. Sci.* 71 (2005) 266–290.
- [28] M. Kearns, Efficient noise-tolerant learning from statistical queries, in: *Proceedings of the 25th Annual ACM Symposium on Theory of Computing*, San Diego, CA, 1993, pp. 392–401.
- [29] M. Kearns, Efficient noise-tolerant learning from statistical queries, *J. ACM* 45 (6) (1998) 983–1006.
- [30] A. Li, J. Bradic, Boosting in the presence of outliers: adaptive classification with non-convex loss functions, *J. Am. Stat. Assoc.* 113 (522) (2018) 660–674.
- [31] X. Li, Y. Zhuang, Y. Fu, X. He, A trust-aware random walk model for return propensity estimation and consumer anomaly scoring in online shopping, *Sci. China Inf. Sci.* 62 (2019) 52101.
- [32] T. Liu, D. Tao, Classification with noisy labels by importance reweighting, *IEEE Trans. Pattern Anal. Mach. Intell.* 38 (3) (2016) 447–461.
- [33] P. Long, R. Servedio, Random classification noise defeats all convex potential boosters, *Mach. Learn.* 78 (3) (2010) 287–304.
- [34] N. Manwani, P. Sastry, Noise tolerance under risk minimization, *IEEE Trans. Cybern.* 43 (3) (2013) 1146–1151.
- [35] H. Masnadi-Shirazi, N. Vasconcelos, On the Design of Loss Functions for Classification: Theory, Robustness to Outliers, and Savageboost, *Advances in Neural Information Processing Systems*, vol. 22, MIT Press, Cambridge, MA, 2009, pp. 1049–1056.
- [36] A. Menon, B. Rooyen, C. Ong, B. Williamson, Learning from corrupted binary labels via class-probability estimation, in: *Proceedings of the 32nd International Conference on Machine Learning*, Lille, France, 2015, pp. 125–134.
- [37] N. Natarajan, I. Dhillon, P. Ravikumar, A. Tewari, Learning with Noisy Labels, *Advances in Neural Information Processing Systems*, vol. 26, MIT Press, Cambridge, MA, 2013, pp. 1196–1204.
- [38] D. Nettleton, A. Orriolspuig, A. Fornells, A study of the effect of different types of noise on the precision of supervised learning techniques, *Artif. Intell. Rev.* 33 (4) (2006) 275–306.
- [39] G. Patrini, F. Nielsen, R. Nock, M. Carioni, Loss factorization, weakly supervised learning and label noise robustness, in: *Proceedings of the 33rd International Conference on Machine Learning*, New York, NY, 2016, pp. 708–717.
- [40] G. Pisier, *The Volume of Convex Bodies and Banach Space Geometry*, Cambridge University Press, 1989.
- [41] H. Ramaswamy, C. Scott, A. Tewari, Mixture proportion estimation via kernel embeddings of distributions, in: *Proceedings of the 33rd International Conference on Machine Learning*, New York, NY, 2016, pp. 2052–2060.
- [42] U. Rebapragada, C. Brodley, Class noise mitigation through instance weighting, in: *Proceedings of the 18th European Conference on Machine Learning*, Warsaw, Poland, 2007, pp. 708–715.
- [43] B. Rooyen, A. Menon, R. Williamson, Learning with Symmetric Label Noise: The Importance of Being Unhinged, *Advances in Neural Information Processing Systems*, vol. 28, MIT Press, Cambridge, MA, 2015, pp. 10–18.
- [44] C. Scott, G. Blanchard, G. Handy, Classification with asymmetric label noise: consistency and maximal denoising, in: *Proceedings of the 26th Annual Conference on Computational Learning Theory*, Princeton, NJ, 2013, pp. 489–511.
- [45] R. Wang, T. Liu, D. Tao, Multiclass learning with partially corrupted labels, *IEEE Trans. Neural Netw. Learn. Syst.* 99 (2017) 1–13.
- [46] H. Xu, C. Caramanis, S. Mannor, Robustness and regularization of support vector machines, *J. Mach. Learn. Res.* 10 (2009) 1485–1510.
- [47] L. Xu, K. Crammer, D. Schuurmans, Robust support vector machine training via convex outlier ablation, in: *Proceedings of the 21st Conference on Artificial Intelligence*, Boston, MA, 2006, pp. 536–542.
- [48] H. Xue, H. Xu, X. Chen, Y. Wang, A primal perspective for indefinite kernel SVM problem, *Front. Comput. Sci.* 14 (2) (2020) 349–363.
- [49] Z.-H. Zhou, Abductive learning: towards bridging machine learning and logical reasoning, *Sci. China Inf. Sci.* 62 (2019) 076101.